FeeBL preliminary manual: beam-beam simulation, collider benchmarks and monochromatization

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Abstract

The collision energy, beam polarization, and electromagnetic deflection of particles in an electron-positron collision affect precision measurements of luminosity and cross sections. For these reasons, and more, we present a new tool, Fast $e^{\pm}e^{\pm}$ Beam and Luminosity (FeeBL) and compare its predictions with published LEP, Future Circular Collider (FCC), International Linear Collider (ILC) and Hybrid Asymmetric Linear Higgs Factory (HALHF) calculations. We identify and correct errors found in other beam-beam simulators; the beam pair implementation is improved by adding finite-mass equivalent-photon and lowest-order matrix-element models. LEP beam electromagnetic deflection differs by approximately 2.5% from published boundary means, plausible and within uncertainty. Beam depolarization is benchmarked with known ILC depolarization results and are plausible relative to the published spread. We have also implemented horizontal-dispersion monochromatization and tested it at a nominal collision energy of 125 GeV. Selected linear-collider inputs give observed peak full widths of approximately 1 MeV, at an instantaneous luminosity that is roughly 1% of nominal. These are exploratory collision simulations, not demonstrated accelerator optics or Higgs width sensitivities. An FCC-ee scan for monochromatization fails to reach full widths comparable to Higgs width, even at 175 mm dispersion. Installation, input cards, the graphical user interface (GUI) viewer, and reusable benchmark data are documented.

Contents

1 Scope of simulation **2**

2 Installation, input cards and output **6**

3 Reference comparisons and uncertainty treatment **13**

4 Depolarization and independent spin tests **27**

5 Horizontal-dispersion monochromatization **32**

6 Program and data release **41**

7 Acknowledgments **41**

1 Scope of simulation

In e^+e^- colliders, intense electron and positron bunches and fields focus and disrupt each other. The fields cause beamstrahlung, secondary-particle production, spin precession, and spin-flip transitions. The fields also deflect out-going particles, be they from the bunches or physics processes like Bhabha scattering, at levels that may influence observables. For example, this altered the small-angle Bhabha rate used for luminosity determination at LEP [1]. At a collider with polarized beams, the luminosity-weighted polarization also affects polarized cross sections. The beam energy spread of the beams and the collision energy can affect threshold or resonance scans. For reasons like these, there is a need at e^+e^- collider experiments for fast and precise beam and beam-beam simulations.

Fast $e^{\pm}e^{\pm}$ Beam and Luminosity (FeeBL) extends upon the Guinea-Pig++ code lineage and retains the original authors' attribution [2, 3]. Beam-beam simulations are implemented by representing each bunch by weighted macroparticles in longitudinal field slices. Successive slice encounters use transverse meshes and a fast Fourier transform field solver; additional transverse meshes extend field evaluation beyond the central mesh. The simulations in this paper used both a multi-threaded CPU implementation as well as a single GPU implementation and fixed meshes. Adaptive meshes and multiple GPU implementations are still currently in development. Simulation settings, such as radiation and secondary processes, are enabled individually by an input card. The physical approximations include locally constant fields approximation (LCFA) for the radiation kernel and, by default, collinear photon emission. A complete accelerator lattice, ring equilibrium, and detector response are outside this collision model and must currently be handled by other programs.

Compton secondaries use the unpolarized Klein-Nishina distribution for a photon antiparallel to the primary particle direction [4]. The final electron and photon retain the incident photon energy, conserve four-momentum, and satisfy their mass shells. Backward electron recoil is allowed. The collision driver retains its equivalent-photon flux and virtuality cuts, so this is a head-on real-photon kernel within a quasi-real photon approximation, not an arbitrary-angle or exact off-shell calculation.

The comparisons use published Guinea-Pig, Guinea-Pig++, Cain and independent field calculations. These references are simulation predictions rather than exact measurements of an underlying beam interaction. Where original particle records are not supplied, we reconstruct Gaussian bunches from published moments. This permits useful consistency tests, but cannot reproduce unknown energy tails, angular correlations, or imported beam structure. Agreement and numerical convergence are therefore assessed separately.

The study of monochromatization is restricted to horizontal dispersion. Vertical dispersion, longitudinal correlations, and chromatic waist matching are separate future studies, not mechanisms validated by the results here. For transparency, the colliders studies here are the International Linear Collider (ILC), the Linear Collider Facility (LCF), the Hybrid Asymmetric Linear Higgs Factory (HALHF), and the electron-positron Future Circular Collider (FCC-ee).

Table 1: Benchmarks and the subject that each attempts to address.

Family	Setting	Principal observables
LEP	Z-pole bunches	Bhabha deflection, acceptance bias
ILC	500 GeV	Incoherent pair yields and luminosity weighted polarization and spin tracking
ILC	1 TeV	Full luminosity spectra modeling
HALHF	250, 380, 550 GeV	Full luminosity spectra and beam pair modeling
Horizontal dispersion	125 GeV	Collision energy peak width and instantaneous luminosity changes for ILC/LCF, HALHF and FCC-ee

1.1 Notation, coordinates, units and observables

We use invariant mass as W as well as collision energy, W_0, and center-of-mass energy, $\sqrt{s}$, in this document and an attempt has been made to ensure that they are not used interchangeably within one context to ensure minimal confusion. We use natural units in kinematic equations but retain explicit units in inputs and tables. The longitudinal axis is $+z$; the two incoming beams travel along $+z$ and $-z$. Beam 1 and beam 2 are identified explicitly in each input card. A 'local angle' about a beam direction is distinct from an angle about the common $+z$ axis and is noted when used. For beams with crossing angles, the full crossing angle is θ_c, so equal axis beam energies of E_0 give $W_0 = 2E_0\cos(\theta_c/2)$ in the ultrarelativistic limit. For unequal beams, like HALHF, with beam energies of E_{01} and E_{02} we find $W_0 = 2\sqrt{E_{01}E_{02}}\cos(\theta_c/2)$.

By the normalized emittance convention the transverse size at a waist follows as

$$\sigma_{x,y}^2 = \epsilon_{n,x,y}\beta_{x,y}/\gamma. \tag{1}$$

Published emittance and beta function values from references are retained when rounded tabulated sizes differ from this relation. The bunch population is a number of physical particles; the macroparticle count is separate and only determines sampling. Their ratio sets a macroparticle weight, but generally we will not mention this distinction so it is noted here. Similarly, counts and yields are not true physical counts and yields and must be scaled by their weights. For example, when comparisons of different runs of yields are compared later, they are normalized by their weights to ensure comparisons are unweighted and residuals are not sensitive to differences in weights and are only sensitive to what is being investigated.

For positive collision weights w_k,

$$\mathcal{L}_{\mathrm{BX}} = \sum_k w_k, \qquad \mathcal{L} = f_{\mathrm{BX}}\mathcal{L}_{\mathrm{BX}}. \tag{2}$$

When the bunch crossing luminosity, $\mathcal{L}_{\mathrm{BX}}$, is in m^{-2}, the reported instantaneous luminosity in $\mathrm{cm}^{-2}\mathrm{s}^{-1}$ is $10^{-4}f_{\mathrm{BX}}\mathcal{L}_{\mathrm{BX}}$. Here $f_{\mathrm{BX}} = f_{\mathrm{rep}}n_b$ counts bunch crossings per second, which may not be equivalent to accelerator crossings under some circumstances. Beam pair counts are per crossing and per weight; reference production comparisons divide them by the matching electron-positron luminosity to obtain effective daughter-production cross sections. The high-energy, peak region, selections of luminosity spectra used here are $W \geq 0.98W_0$ or $W \geq 0.99W_0$, with no upper bound. These selections are energy windows, not the highest two or one percent of events ranked by energy.

The peak response, complete-spectrum RMS, and observed full width at half maximum (FWHM) are also distinct metrics used for luminosity spectra. Radiation tails can dominate the luminosity spectra RMS even when the peak is narrow. We use the beta convolved Gaussian response model in Sec. 5 when its extraction passes the stated numerical checks, and label a directly measured half-height width as an observed FWHM.

For spin expectation $\boldsymbol{s}$ and particle direction $\widehat{\boldsymbol{p}}$, the local helicity is $h = \boldsymbol{s}\cdot\widehat{\boldsymbol{p}}$. Positive helicity denotes something that is right hand dominant, $P = (N_R - N_L)/(N_R + N_L)$. For an accepted collision sample A, we write

$$P_{-,A} = \langle h_- \rangle_A, \quad P_{+,A} = \langle h_+ \rangle_A. \tag{3}$$

We note that these expectation values are weighted, which means, in this context, luminosity weighted. We quote polarization changes as percentages of unit polarization: an absolute change of 0.001 is written as 0.1%. A relative loss instead divides the change by the initial polarization and is identified explicitly. The spin benchmarks start with unit spin along each beam's nominal propagation axis.

For comparison with the explicitly defined 2009 result [5], let $q_1 = s_{z,1}$ and $q_2 = -s_{z,2}$, and define

$$\Delta P_{z,L} = 100\left[1 - \sqrt{\langle q_1 q_2 \rangle_L}\right]\%. \tag{4}$$

Here the numerical value in brackets is expressed in percent by the factor 100; $\langle\cdot\rangle_L$ denotes the inclusive luminosity average. Replacing q_i by h_i defines the local-helicity product loss. Neither is the conventional effective polarization of a left–right asymmetry. For annihilation with only the LR and RL initial channels, the luminosity-averaged coefficient is [6]

$$P_{\mathrm{eff},A} = \frac{\langle h_- \rangle_A - \langle h_+ \rangle_A}{1 - \langle h_- h_+ \rangle_A}. \tag{5}$$

It is a ratio of averaged cross-section coefficients. Replacing the product average by the product of separate averages removes collision correlations. Process-dependent kinematic weights may be required. In the ideal fully polarized limit of this spin benchmark, $P_- = P_+ = 1$ and the LR/RL effective polarization is 0/0. The tiny initial helicity reduction from finite emittance does not make this a useful P_{eff} benchmark. Outgoing whole-bunch and downstream polarimeter averages require their own definitions [7].

1.2 Pair production and corrections to the inherited model

Beam-induced electrons and positrons contribute to detector occupancy over a wide range of transverse momentum and polar angle. Their prediction requires both the production rate and the differential distribution to be correct before transport through the bunch fields. We distinguish the Breit–Wheeler channel, $\gamma\gamma \to e^+e^-$, the Bethe–Heitler channel, $\gamma e^\pm \to e^\pm e^+ e^-$, and the Landau–Lifshitz channel, $e^+e^- \to e^+e^-e^+e^-$. These names follow the beam-background convention of Ref. [8]; the Bethe–Heitler target is a beam lepton.

We inspected the inherited GUINEA-PIG++ implementation and compared individual formulae and samplers with independent QED calculations. Table 2 summarizes verified implementation defects and a change in population bookkeeping. This attribution is specific to that source revision and does not establish which revisions or options were used for every published collider prediction. The distributed source audit records the upstream revision and file hashes.

Table 2: Verified implementation defects and population-bookkeeping changes. The angular-sampling, dispatch and spectator-counting changes apply to pair modes 1 and 2. The normalization correction applies to the shared Breit–Wheeler kernel; exact Bethe–Heitler and Landau–Lifshitz amplitudes carry their own normalization. In mode 2 the inexpensive Breit–Wheeler estimate controls sampling effort, with a compensating weight. The listed numerical effects isolate individual components; they are not predictions for the net change in a collider background.

Component	Inherited behaviour	Corrected behaviour
Born normalization	The literal prefactor is 1.319% below $\pi r_e^2/2$.	Construct the prefactor from the classical lepton radius and the numerical value of π.
Forward angle sampling	Absolute inverse-CDF tolerance truncates the small-angle distribution near $1 - \lvert\cos\theta^*\rvert \simeq 2^{-20}$.	Sample a logarithmic angular variable with rejection and stable endpoint expressions.
Landau–Lifshitz dispatch	Disabling both real-photon generation and loading also skips virtual-photon collisions.	Dispatch the channel independently; the input must still enable its virtual-photon population.
Scattered beam leptons	With `beam_pair=1`, the same counters include scattered beam leptons. Expected Bethe-Heitler and Landau-Lifshitz counts are larger by 3/2× and 2× respectively.	Retain scattered beam leptons separately from the two produced daughters.

Including scattered beam leptons is intentional; the distinction matters when those counters are interpreted as produced-daughter yields.

The Born cross section vanishes for $\hat{s} \leq 4m_e^2$. This is a pair-production threshold; each daughter separately satisfies $E \geq m_e$. A higher production-energy selection such as $E > 5\,\mathrm{MeV}$ is an analysis choice. The corrected implementation evaluates the threshold before angular sampling and uses stable logarithms near the forward endpoints.

FeeBL exposes three pair models. Mode 0 retains the EPA production model with the normalization and threshold corrections already present in FeeBL; it does not reproduce the historical code byte for byte. Mode 1 adds the stable forward-angle sampler and the finite-mass term of the transverse equivalent-photon flux [9],

$$dn_T = \frac{\alpha}{\pi}\frac{dx}{x}\frac{dQ^2}{Q^2}\left[\frac{1+(1-x)^2}{2} - (1-x)\frac{Q^2_{\min}}{Q^2}\right], \qquad Q^2_{\min} = \frac{m_e^2 x^2}{1-x}. \tag{6}$$

Here x is the photon energy fraction and $Q^2 = -q^2 > 0$. Omitting the second term is a leading-logarithmic approximation, distinct from the normalization error and population definitions in Table 2. The correction weights the pair channel without silently changing the flux used by other background channels.

Mode 2 evaluates the complete lowest-order QED matrix elements for the Bethe–Heitler and Landau–Lifshitz final states, including identical-lepton exchange. Exact internal four-momenta and phase-space weights replace the on-shell $\gamma\gamma$ approximation and its hard `pair_q2` veto. The calculation uses importance sampling: if $g(\Phi)$ is the proposal density, the contribution is proportional to $\overline{|\mathcal{M}(\Phi)|^2}\,d\Phi/g(\Phi)$. The support and precision of g therefore require the same care as the matrix element. A large, rare weight must be diagnosed rather than silently discarded. The current host-code mapping to the beam directions still uses a common transverse boost; exact internal kinematics do not remove that angular approximation. These are tree-level calculations, not a complete radiatively corrected prediction.

The symbolic calculations use Mathematica [10], FeynCalc 10.2.1 [11, 12] and FeynHelpers [13], with diagrams generated by FeynArts 3.12 [14]. The distributed scripts retain the amplitude construction and high-precision fixed-point comparisons. Ward identities, Lorentz transformations and identical-particle exchange test the amplitudes; angular distributions, phase-space weights and collision ensembles test different parts of the calculation. Agreement of a matrix element at fixed phase-space points alone does not validate a simulated collider yield. The secondary tracker retains double-precision Cartesian momentum and derives its energy from the mass shell. This removes the feedback of rounded velocity components into the energy during subsequent steps. Magnetic transport does no work; radiative recoil is checked against the available kinetic energy. The production kernels and transport use the same lepton-mass constants. Prescribed near-rest reversal, weak-field, magnetic no-work and recoil controls pass in both strict and production arithmetic. The complete regression run passes 105 native targets, including eleven required QED and transport targets. Seven serial-CPU collision pilots also complete. The HALHF 550 GeV configuration that exposed the mass-shell failure now finishes with its original input card, seed and 0.5 MeV daughter-admission setting. Its mixed secondary population and mesh-coverage warning are retained; this replay does not establish a converged pair yield. The separate strict adaptive-field qualification remains unmet. Additional beam-background and pair channels remain development components whose validation is work in progress.

1.3 Secondary-particle observables

Secondary yields below count individual electrons plus positrons: one pair vertex supplies two daughters. A production-energy threshold and an energy threshold after transport select different populations. For a particle with momentum $\boldsymbol{p}$,

$$p_T = \sqrt{p_x^2 + p_y^2}, \qquad \theta_{\mathrm{near}} = \mathrm{atan2}(p_T, |p_z|). \tag{7}$$

The nearest-axis angle folds both beam directions into $[0, \pi/2]$. The angle about the common $+z$ axis instead uses p_z without its absolute value and spans $[0, \pi]$. A backward population can form a

narrow vertical feature near $\log_{10}\pi$ without any coordinate error. Selection by spatial $z > 0$ or $z < 0$ is separate from the sign of p_z.

2 Installation, input cards and output

The accompanying repository contains the simulator, analysis tools, example cards, digitized reference data and this paper. The benchmark results were produced with the source revision recorded in each result manifest; the publication package preserves those identities separately from subsequent documentation changes. Replotting stored results and executing a new collision are separate operations. A successful program exit establishes completion, not agreement with a reference or numerical convergence.

2.1 Build and first calculation

The supported minimal build uses Linux or Linux under Windows Subsystem for Linux, CMake 3.16 or newer, a C++17 compiler and FFTW3. Python is used for benchmark orchestration and analysis. The runner requires Python 3.10 or newer; replotting requires NumPy 1.24 or newer and Matplotlib 3.7 or newer. SciPy is required for viewer finalization as well as numerical fitting. The browser viewer is a static HTML application. Optional CUDA and parallel backends have separate requirements and are not needed to reproduce the serial calculations discussed here.

From the repository root, a standard CPU build is

```
sudo apt-get install build-essential cmake libfftw3-dev \
  libhdf5-dev python3 python3-venv nodejs
python3 -m venv .venv
. .venv/bin/activate
python -m pip install -r requirements-analysis.txt
cmake -S . -B build -G "Unix Makefiles" \
  -DCMAKE_BUILD_TYPE=Release -DFEEBL_ENABLE_CUDA=OFF
cmake --build build --parallel 4
ctest --test-dir build -LE strict --output-on-failure
ctest --test-dir build -L strict --output-on-failure
```

The first test command runs software regressions. The second runs the experimental adaptive-grid accuracy gate, whose failure remains explicit. Its round-Gaussian field error of 5.14% is also present in the uniform-grid control; it cannot be attributed to adaptive interfaces alone. The interface potential and observed-order requirements also fail. Excluding this gate does not qualify the adaptive solver. Preserve both test reports. The executable is `build/src/feebl`. For an FFTW installation outside the system search path, provide its installation prefix with `-DCMAKE_PREFIX_PATH=/path/to/fftw`. The CMake configuration output identifies the selected library. Keep that output and the compiler version with a new benchmark. The Makefiles generator is specified because the existing spin-test link procedure uses its generated link information.

CPU worker pools are included in the normal build and are selected by the input card, not by the number of compilation jobs. OpenMP is a separate build option, `-DFEEBL_ENABLE_OPENMP=ON`. CUDA support requires `-DFEEBL_ENABLE_CUDA=ON`, a CUDA toolkit and a supported GPU. The resident GPU modes impose different process and mesh restrictions; `doc/PARALLEL_INPUT_OPTIONS.md` specifies each contract. An unavailable explicit backend is rejected unless fallback is explicitly authorized. The collider results here use the serial backend. Compilation of an accelerated backend does not by itself establish numerical equivalence or speedup for a particular physics card.

The packaged runner lists the available cases and creates a separate output directory for a small first collision:

```
python benchmarks/run.py list
```

```
python benchmarks/run.py run smoke \
  --feebl build/src/feebl --output runs/smoke
python benchmarks/run.py verify
```

The smoke case checks installation and the output path. Its small numerical settings are not intended to reproduce a precision collider benchmark. The benchmark manifest lists the native cases and reference numerical datasets. The manifest identifies the input card, accelerator and parameter blocks, supporting files, and stored reference outputs for each full example. The runner copies the required inputs before executing the selected blocks. It does not silently substitute another accelerator configuration.

To regenerate the manuscript figures and numerical tables from preserved native reductions and reference ordinates without rerunning the collision solver, combine the preserved comparison set with the completed ensemble comparisons, then run the plotting command:

```
mkdir -p results
mkdir results/paper_comparisons
cp benchmarks/example_output/comparisons/* \
  results/paper_comparisons/
cp benchmarks/example_output/seed_calibration_partial/comparisons/* \
  results/paper_comparisons/
python benchmarks/analysis/regenerate_paper.py \
  --comparisons results/paper_comparisons \
  --output results/paper_replay
```

The second copy replaces the ILC spectrum and geometry comparisons with their complete historical ensembles. The plotting command separately requires the packaged, verified 60-collision reduction for current LEP focusing and ILC/HALHF pair comparisons. Historical HALHF luminosity spectra remain distinct from validation of the revised pair kernels. The output includes a generation receipt with input identities. The included numerical values remain the results of their recorded calculations. A new run should be compared at matched physical settings, binning and cuts before its output replaces a saved result.

2.2 Input-card structure

The native command selects an accelerator block and a parameter block from `acc.dat` in the working directory:

```
feebl --make_gui_files <accelerator_block> <parameter_block> <output_file>
```

The accelerator block defines the bunches and optics; the parameter block selects numerical resolution, processes and outputs. For complete runnable cards, use `benchmarks/cases/`. The following excerpt illustrates the organization and the opposite dispersion signs; it is not a complete card:

```
$ACCELERATOR::example
{
  energy=62.501531;
  particles=2.0;
  beta_x=3.25;
  beta_y=0.41;
  sigma_z=300.0;
}
$PARAMETERS::example_settings
{
  ip_dispersion.1=0.0003712;
  ip_dispersion.2=-0.0003712;
  incoming_map_reslice=1;
}
```

Native cards retain historical units. In this example, energy is in GeV, population is in units of 10^{10}, beta functions are in mm, bunch length is in μm and dispersion is in metres. The inconsistent length scales are interface conventions, so every edited quantity should be checked against the generated `docs/INPUT_REFERENCE.md`. A suffix `.1` or `.2` sets the corresponding beam; an unsuffixed assignment can set both. In particular, oppositely directed beam maps often require different signs.

Table 3: Input groups relevant to the examples. Complete cards determine all process flags and retain the exact native keyword spellings and units.

Group	Interpretation and common error to avoid
Energy, population, emittance and beta	Physical incoming bunch parameters. Macroparticle count changes sampling, not the requested physical population.
Energy spread	Applied to primary particles before incoming maps. Enabling dispersion for a monoenergetic beam does not create the intended energy–position correlation.
Crossing, crab and waist settings	Geometric crossing, RF-like bunch tilt and crab-waist optics are distinct. Match signs and map order for both beams.
`ip_dispersion.1`, `ip_dispersion.2`	Horizontal IP dispersions in m. The canonical map changes both x and the longitudinal coordinate.
`incoming_map_reslice`	Reassigns slices after canonical incoming maps. Required by the high-dispersion examples to retain scheduling consistency.
Particle, mesh, slice and time settings	Numerical controls. Increase each separately at fixed physical inputs before a combined refinement.
Spectrum limits and bins	Choose support and peak resolution before running. A narrow peak cannot be recovered from a single broad energy bin.
Radiation, spin and secondary switches	Define the physical model. Switching one process can change subsequent random draws and trajectories.

For horizontal dispersion, the canonical map is

$$x \longmapsto x + D\delta, \qquad \zeta \longmapsto \zeta + Dp_x, \tag{8}$$

where p_x is normalized canonical transverse momentum and $\delta = (E - E_0)/E_0$. The longitudinal term is required for symplecticity. Dispersion precedes the crab transformation, including its companion energy kick. Changing their order is a physical change, not a formatting choice. After these maps, `incoming_map_reslice=1` reassigns particles using their transformed longitudinal coordinates while preserving identity, spin and weight. A transformed particle outside the declared longitudinal domain causes setup to fail instead of being silently discarded or clipped.

Imported primary beams can use the native ASCII loading interface. A card marked `loaded_beam_is_pretransformed=1` declares that the incoming transformations have already been applied; it skips crossing, roll, offsets and incoming maps. It must not be combined

with automatic incoming-map reslicing. Apply each transformation exactly once. The native imported-beam path should not be confused with auxiliary HDF5 conversion utilities.

Incoherent-pair settings. The models in Sec. 1.2 are selected by `pair_model=0`, 1 or 2. For a pure incoherent-daughter study, a parameter-block excerpt is

```
pair_model=1; do_pairs=1; track_pairs=1;
do_photons=1; do_photon_lumi=1; electron_ratio=1;
beam_pair=0; do_compt=0; do_coherent=0; do_trident=0;
load_events=0; do_bhabhas=0;
store_pairs=2; pair_ratio=1; pair_ecut=0.005;
```

The energy cut is in GeV and acts on each daughter at production. `store_pairs=2` preserves birth records in `pairs0.dat` and tracked final records in `pairs.dat`; these are two stages of the same population, not additive samples. A smaller `pair_ratio` reduces storage and requires its inverse weight for per-crossing yields. Use final admitted records for the comparison without applying the birth cut a second time. Native production-expectation counters precede admission and need not equal the stored daughter count. Complete benchmark cards specify the remaining beam, numerical and photon settings.

2.3 What to preserve from a run

Keep the exact materialized input card, executable and source identity, resolved parameter summary, random seeds, numerical settings, stdout/stderr and complete scientific outputs. A scalar result without these inputs cannot be used to distinguish a physics change from an implementation or numerical change. RNG checkpoint loading and saving are disabled by default. When enabled, the versioned checkpoint stores the complete generator state, including cached Gaussian deviates, and is validated before state changes. An incompatible historical checkpoint is rejected rather than partially read. Preserve the input checkpoint and its hash alongside the card; the seed alone does not specify a resumed stream. See `docs/RNG_CHECKPOINTS.md`. The supplied examples retain compact genuine outputs and comparison tables; they do not require the complete collection of earlier study folders.

The main text summary reports evaluated inputs and integrated results. Native luminosity spectra retain bin edges, weights and flow information. The viewer sidecars contain frame diagnostics, sampled primary coordinates, field maps where enabled, and secondary or imported-particle records. The final manifest records their units, timing, identities and checksums. Checksums establish that a dataset is internally consistent; they do not establish physical correctness. Pair-production counters, stored-particle counts and display samples must retain their separate normalizations.

2.4 Viewer and diagnostic scopes

The viewer loads a finalized local run directory; an input card alone is not a viewer dataset. For the smoke example, finalize its outputs with the actual executable, input card, source and collision-rate convention:

```
python tools/finalize_gui_dataset.py runs/smoke \
  --binary build/src/feebl --card runs/smoke/acc.dat --source .
python tools/serve_viewer.py --run runs/smoke
```

The server prints the local browser address and serves only the selected run and viewer assets. Finalization obtains the full horizontal crossing angle and $f_{\mathrm{rep}} n_b$ from evaluated native inputs and records their provenance. Explicit angle or frequency overrides must describe the actual run. The saved demonstration can be opened without a new simulation:

```
python tools/serve_viewer.py \
  --run benchmarks/example_output/gui_smoke
```

The folder chooser also accepts another finalized directory. The browser reads those selected files locally. The persistent frame slider and playback control remain available while scrolling. A global frame/total control changes the diagnostic scopes together; individual panels can override it and show their active scope explicitly. The beam direction arrows describe the saved display-coordinate convention.

The animation uses a deterministic primary-particle subset for display. Scientific moments and histograms are read from native accumulators, not reconstructed from those visible points. The selected frame synchronizes the plots. Table 4 distinguishes their temporal meanings. The first

Table 4: Frame and total modes in the viewer. A frame interval contains collision contributions since the preceding saved state.

Quantity	Current frame	Total
Luminosity and radiation spectra	Contributions in the frame interval	All collision contributions
Collision energy and polarization	Luminosity-weighted accepted pairs in the interval	Luminosity-weighted accepted pairs over the collision
Beam moments	Unweighted stored primary state	Pooled collision-weighted first and second moments
Secondary kinematics	One stored state per admitted tracked record	Final stored records
Imported-particle deflection	Change since admission at the selected state	Change since admission at the final state

stored frame already follows a collision interval and is not an exact pre-collision snapshot. Frame moments use stored slice states, whereas the displayed primary cloud can drift inactive slices to a common endpoint. These are different sampling conventions. Total RMS values are formed from pooled first and second moments, not from averages of frame RMS values.

Nominal spin settings, whole-bunch helicity and luminosity-weighted electron and positron helicities are shown separately. High-energy means and polarizations use $W \geq 0.98W_0$. The interface does not substitute a spin-product average for the conventional P_{eff} in Eq. (5). A distribution that was not computed is distinguished from a calculated zero yield. Separate electron and positron χ_e histograms count field-kick evaluations, including repeated substeps. The instrumented coherent-photon path supplies χ_γ; ordinary photon propagation does not. Beamstrahlung and pair-energy histograms have their own native production weights and are not histograms of the displayed sample.

The longitudinal field overlays use (z, x) at $y = 0$ and (z, y) at $x = 0$. They show the transverse electric-equivalent magnitude $E_\perp = \sqrt{E_x^2 + E_y^2}$, converted from native GV/nm to displayed GV/m. A single beam uses a Cividis-derived gradient; two partially transparent blue and orange layers retain the source identity and a common scale. Showing both is not a vector sum: two scalar magnitudes cannot determine the net field. Fixed physical map coordinates prevent an apparent contraction from automatic pixel rescaling. The full-bunch field view drifts copied particles to a common endpoint and evaluates a separate diagnostic mesh. It does not modify the tracked particles or establish field values outside the sampled domain. These overlays do not represent a complete laboratory electromagnetic field or demonstrate external laser-field support. Figure 1 shows the field controls and persistent playback in the packaged installation example, followed by frame-resolved numerical diagnostics from a polarized collision. The latter distinguish the configured polarizations, current bunch means and luminosity-weighted collision values.

The produced-particle panel plots $\log_{10}(\theta/\mathrm{rad})$ against $\log_{10}[p_T/(\mathrm{GeV}/c)]$. Its axis convention,

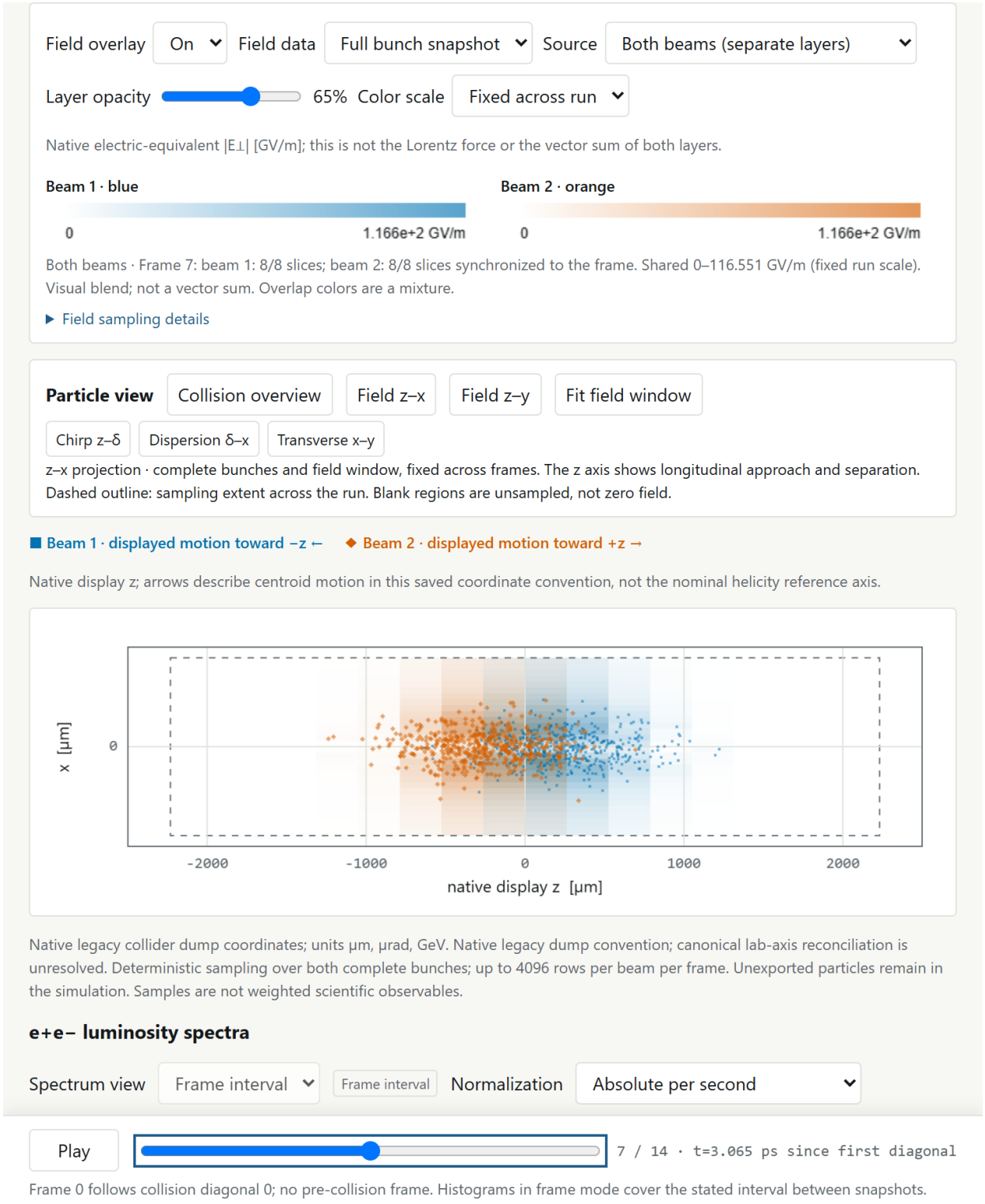


Figure 1: FEEBL viewer at saved frame 7 (numbered 0–14) of the 500 GeV installation example. Blue and orange identify the sampled bunches and their separate transverse electric-equivalent field layers on a common GV/m scale. Overlapping colors do not represent a vector sum. The dashed outline marks the sampling extent. Projection controls and fixed playback permit frame inspection. This small demonstration is not a converged collider prediction.

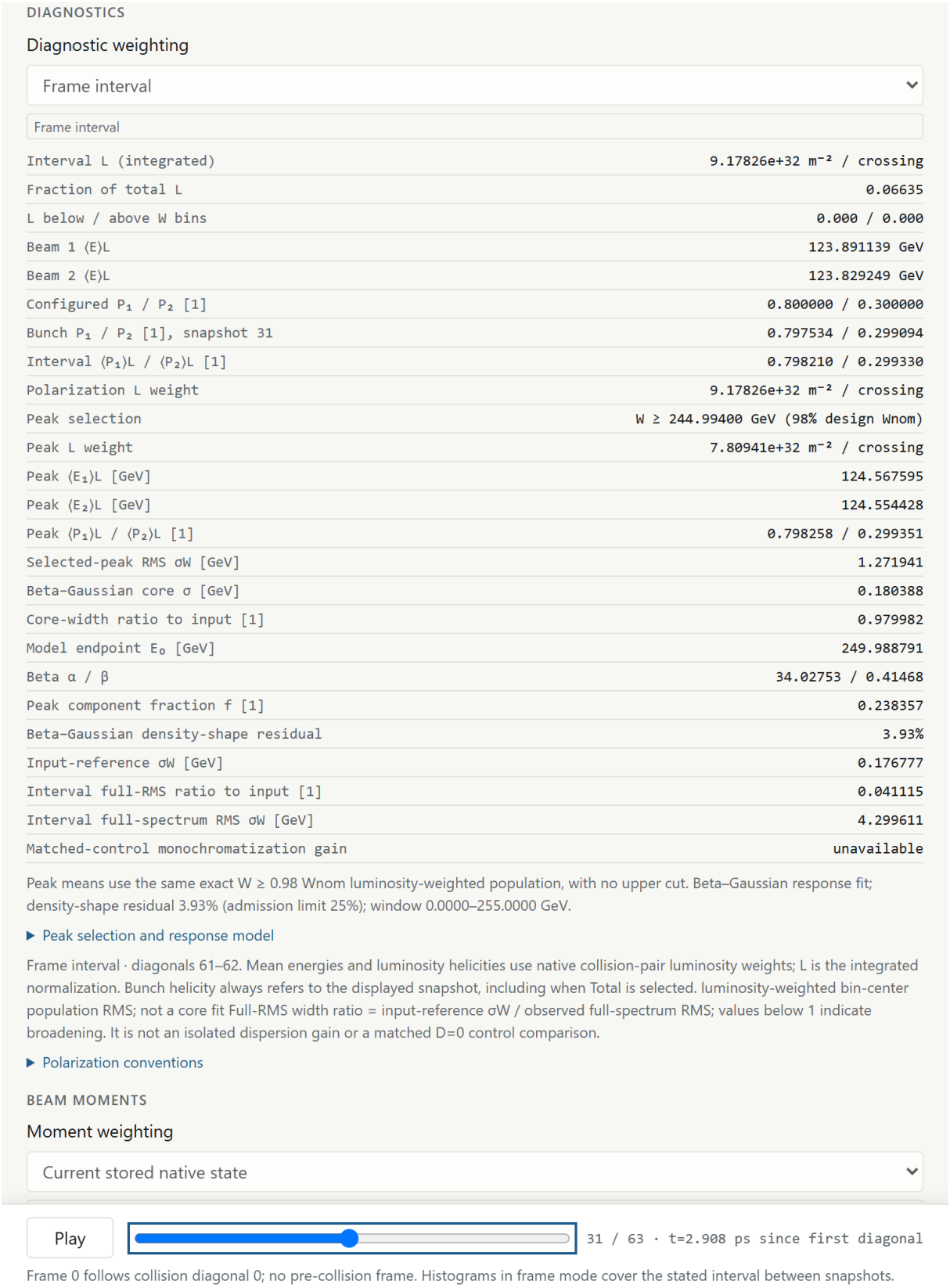


Figure 1: Continued: quantitative diagnostics at frame 31 of a saved 250 GeV polarized collision. The frame-interval selector and playback position remain visible. Energy and polarization means are reported for the interval and its $W \geq 0.98W_0$ subset; the current bunch means use the displayed snapshot. This preserved example illustrates the interface, not a precision validation of the displayed fitted response.

frame/total selector and spatial z selection are independent. Imported deflected particles instead show $\Delta\theta$ and wrapped $\Delta\phi$ relative to their actual native admission directions, joined by event and particle identity. For a two-body Bhabha test, four distributions describe electron and positron polar and azimuthal changes. They are changes since admission, not changes between adjacent saved frames.

For supported imported events, `gui_imported_trajectories.json` stores the initial record identity and the direction at admission, each saved frame and the final state. The finalizer joins persistent particle labels to the event ID, particle slot and PDG ID in `event_identity.json`. It writes `collider_event_deflections.json`, with fixed-bin angular histograms in degrees, frame timing, admission counts and source hashes. The viewer downloads both this analysis and the native records. Event matching and angular extraction therefore require no separate shell sorting script. This adapter currently supports a pure population of ordinary eight-column imported events in the serial, head-on tracker, with at most 8192 identified records and 64 saved frames. Its trajectory rows contain direction and field-gather counts; they do not supply final particle energies or positions.

Plot boxes report means, modes, variances and extrema. Histogram modes are maximum-density bin intervals; binned extrema are occupied-bin edges. Pair summaries refer to physical θ and p_T, not their logarithms. These summaries are not uncertainty estimates. Downloads include the manifest, sampled coordinates, diagnostic JSON, available native files and derived spectrum or particle analysis. The diagnostic JSON retains frame and total moments and polarization. A coordinate CSV is a display sample without all spins or particle identifiers, not a complete beam dump. The original input card remains in the run directory; the viewer has no dedicated complete-run archive or field-map download.

3 Reference comparisons and uncertainty treatment

The packaged reference tables contain values extracted from the primary publications listed in their provenance records. Their original bin boundaries, displayed support and observable definitions are retained. Graphical markers or colored raster cells are not independent simulated particles and cannot determine the reference sample size. We therefore distinguish direct sampling errors from estimated curve uncertainties and numerical sensitivity. The standard error of a mean (SEM) across R independent seeds is $s/\sqrt{R}$, with sample standard deviation s. A three-seed SEM is not a 95% interval. Numerical changes below are controlled sensitivities; they are neither confidence intervals nor calibrated total numerical errors.

Result versions. The focusing and pair figures use the revised implementation and complete independent-collision ensembles. The HALHF luminosity table and spectra retain their original source and executable identities. Those historical results do not qualify the revised pair or Compton kernels; the completed kernel and transport controls have a separate scope.

The production ensembles retain main-grid/source-stencil coverage warnings. The maximum recorded rejection fractions for beam 1/beam 2 are 5.56/4.00% for LEP, 11.1/33.3% for ILC and 53.8/41.2% for HALHF. These count macroparticle visits, not lost physical charge; outer meshes and analytic fallback have separate contributions. Independent-seed errors at a fixed domain do not bound the resulting observable bias. Larger-domain studies at fixed cell spacing remain necessary before precision interpretation.

Table 5 gives the principal physical and numerical inputs for the collider comparisons. Equal beam values are printed once. The complete evaluated input table and figure-to-case mapping accompany the release; these retain energy spreads, emittances, domain sizes, process switches and reproduction status.

For graphical spectra, simulated weights are transferred to the published bins by overlap and

Table 5: Selected collider inputs. N_m is macroparticles per beam and the grid column is $N_x \times N_y \times N_z$. These are simulation settings, not demonstrated convergence thresholds. HALHF energies are asymmetric.

Case	$E_{1,2}$ [GeV]	$N_{1,2}/10^{10}$	β_x [mm]	β_y [mm]	N_m	Grid
HALHF 250 spectrum	375, 41.7	1, 3	3.3	0.1	10000	256×1024×32
HALHF 380	570, 63.3	1, 3	3.3	0.1	10000	128×128×32
HALHF 550	825, 91.7	1, 3	3.3	0.1	10000	128×128×32
ILC 1 TeV	500	1.74	11	0.23	10000	128×128×32
LEP focusing	45.6	12.8	2000	50	40000	128×256×64
ILC pairs	250	2	15	0.4	10000	128×128×32
HALHF 250 pairs	375, 41.7	1, 3	3.3	0.1	10000	128×128×32

both distributions are normalized on the common retained support $\mathcal{I}$. Censored intervals remain gaps. The conditional cumulative-distribution distance is

$$D = \sup_{W \in \mathcal{I}} |F_{\mathrm{sim}}(W \mid \mathcal{I}) - F_{\mathrm{ref}}(W \mid \mathcal{I})|. \tag{9}$$

Full and peak panels are normalized separately. Thus D tests their shapes, not luminosity or probability outside the displayed interval.

3.1 Pulls and estimated reference errors

For bin contents s_i, r_i with matched definitions, the displayed pull is

$$p_i = \frac{s_i - r_i}{\sqrt{(C_s + C_r - C_{sr} - C_{sr}^{\mathsf{T}})_{ii}}}. \tag{10}$$

Both simulation and reference covariance enter. $C_{sr} = 0$ is assumed for independent published-reference comparisons; paired optical changes use the covariance of their within-seed differences. Holding the estimated denominator fixed gives a propagated uncertainty of one on each pull. All lower-panel error bars are consequently ± 1, with endcaps. This display convention does not prove that pulls with estimated variances are standard normal.

For an independent Poisson count N, the absolute error is approximately $\sqrt{N}$ and the relative error is $1/\sqrt{N}$. A weighted count has variance $\sum w_k^2$ under independent Poisson sampling. Normalized bins share a normalization covariance; pair daughters and repeated encounters also share physical histories. The present estimates account for the known sampling structure as follows. Each new LEP realization includes an independently sampled primary field and correlated probe events. Each pair realization includes the fields, production events and correlated importance-sampling copies. For complete independently seeded collisions, the mean and its sampling covariance are

$$\overline{\boldsymbol{y}} = \frac{1}{R} \sum_{a=1}^{R} \boldsymbol{y}^{(a)}, \qquad C_{\overline{y}} = \frac{1}{R(R-1)} \sum_{a=1}^{R} (\boldsymbol{y}^{(a)} - \overline{\boldsymbol{y}})(\boldsymbol{y}^{(a)} - \overline{\boldsymbol{y}})^{\mathsf{T}}. \tag{11}$$

Unit-normalized spectra are normalized on their retained support before averaging. Pair-production comparisons use the ratio of summed daughter counts to summed electron–positron luminosity. Their uncertainty propagates the joint count/luminosity covariance, including its cross terms. Raw per-crossing counts remain separate detector-background diagnostics. The covariance of one collision is $RC_{\overline{y}}$. Repeated encounters and collective fields are correlated; their counts are not substituted for the number of independent collisions. Complete runs vary both the primary bunch realization and secondary or probe sampling. Their scatter therefore includes both contributions without adding a second within-run counting error to the same ensemble covariance. Increasing statistics should reduce the uncertainty of the mean. It can expose a stable discrepancy rather than necessarily reduce its pull.

Bins empty in every replica retain zero measured content. The retained spectrum products use a flagged half-observation display regularizer based on encounter-Kish exposure, projected onto fixed-total space. The new focusing and pair ensembles use their measured collision covariance without this extra term. Zero empirical variance does not establish an absent tail. For the published pair yields, we use an explicitly assumed one-equivalent-crossing variance $2Y_i$; alternatives Y_i and $4Y_i$ test this scale. These are not retrospective estimates of the source authors' actual event count. For cross sections, this variance is divided by the square of the fixed published luminosity; its unreported uncertainty is not inferred. Distinct subprocess categories are never smoothed against one another.

Reference-curve fluctuations are estimated before interpolation or normalization by removing local linear trends [15]. For equally spaced ordinates, a unit-norm three-point contrast is

$$r_i = \frac{y_{i-1} - 2y_i + y_{i+1}}{\sqrt{6}}, \qquad \widehat{v}_i^{\mathrm{raw}} = r_i^2. \tag{12}$$

For unequal spacing, the coefficients are orthogonal to the constant and linear terms at the actual abscissae and have unit squared norm. Spectra are first expressed as densities. A three-point local pilot supplies heteroscedastic scaling; the nearest 15 squared contrast probes in each contiguous interval are pooled using a median with its Gaussian-noise calibration. A five-point line fit and 9/23-probe pooling are companion checks. Curvature, extraction and plotting can contribute to the estimate. No displayed ordinate is smoothed. Gaps are not bridged, and short support segments borrow a curve-wide fluctuation scale.

The variance floor is the larger of the local fluctuation variance and a uniform graphical-reading variance (half-width squared divided by three). Reading half-widths are $0.1\,\mu$rad for azimuthal focusing, 1% of the peak for its histogram, 3% of the ordinate for HALHF, and 0.008 dex or 0.6 in dN/dx for the full or peak ILC spectrum. These are extraction assumptions. The normalization Jacobian $J_{ij} = (\delta_{ij} - p_i)/\sum m_k$, for $p_i = m_i/\sum m_k$, propagates the resulting bin-mass covariance. Shared normalization therefore produces off-diagonal terms even when original ordinate errors are assumed independent.

The covariance check reserves eight collisions for training and four independent collisions for validation. Differences of these two means use $S_{\mathrm{train}}(1/8 + 1/4)$; under Gaussian sampling their standardized values have a t_7 reference with expected RMS $\sqrt{7/5}$, rather than unity. Three predetermined force-free observables separately compare mean energy, a second moment and a high-energy integral with their analytic predictions. Bin coverage and pull RMS remain descriptive because bins are correlated. These checks neither calibrate the published-curve uncertainty nor establish spatial or time convergence. Full covariance matrices are retained, but their finite-ensemble rank and the estimated reference errors preclude an unqualified global chi-square significance.

The completed 12-seed geometry controls give held-out RMS values 1.02–1.07 and pointwise 95% coverage of 97.7–98.1%. Their three predetermined scalar observables have $|t_{11}| \leq 1.47$ against the analytic predictions. The ILC 1 TeV full and peak spectra give RMS values 1.29 and 0.92, with coverage 94.9% and 97.6%, respectively. These checks support the measured sampling scale at the tested resolution; four held-out collisions do not establish precision coverage in every bin.

3.2 LEP Bhabha focusing

The independent-field ensemble uses ten primary-bunch realizations, each with 1024 collision vertices at each initial polar angle, 31.3 and 51.6 mrad. Each event contains an electron and a positron, giving 40 960 tracked particles in total. The boundary cohorts share their vertex and time draws, and 48 initial-azimuth strata receive equal weight. Gaussian longitudinal overlap and hourglass dependence are retained. The initial transverse boost of Ref. [1] is omitted, so these probes approximate the focusing calculation rather than a complete radiative Bhabha sample. The identified-event interface in Sec. 2 exports native initial and tracked directions as JSON, then

produces frame-resolved and final angular distributions automatically. This replaces the separate output-sorting step described for the GUINEA-PIG++ workflow in Ref. [16, Sec. 4.3.1.1]; event identity and coordinate conventions accompany the numerical output.

For focusing $\Delta\theta = \theta_i - \theta_f$ and an angular rate proportional to θ^{-3}, the leading acceptance correction is

$$\frac{\Delta\mathcal{L}}{\mathcal{L}} = -\frac{2}{\theta_{\min}^{-2} - \theta_{\max}^{-2}} \left[\frac{\langle \Delta\theta \rangle_{\min}}{\theta_{\min}^{3}} - \frac{\langle \Delta\theta \rangle_{\max}}{\theta_{\max}^{3}} \right]. \tag{13}$$

Each field supplies the azimuthal means, a conditional focusing histogram and both boundary means. Their covariance gives the standard error of the ensemble mean and the uncertainty of Eq. (13). Two matched zero-field controls check transport separately; they are not subtracted from unrelated fields. Sampling errors exclude input-model and numerical resolution uncertainties.

Table 6: LEP electron-probe focusing and the leading boundary estimate of the luminosity bias. FEEBL values are means of ten independent fields with standard errors; differences are relative to the reference. The reference bias applies Eq. (13) to the published boundary means rather than using the distinct full-acceptance result.

Observable	Reference [1]	FEEBL	Difference
Inner focusing [μrad]	12.81	13.123 ± 0.079	+2.45%
Outer focusing [μrad]	11.19	11.490 ± 0.062	+2.68%
Bias [%]	-0.10425	-0.10675 ± 0.00066	+2.39%

Applying Eq. (13) to the published boundary means gives -0.10425%. The published simulation's -0.1059% acceptance result uses its own event/acceptance construction; it is retained as context, not inserted into this like-for-like estimator comparison. The simulated boundary means are $13.123 \pm 0.079\,\mu$rad and $11.490 \pm 0.062\,\mu$rad, respectively 2.45% and 2.68% above the reference. Their correlated boundary estimate is $(-0.10675 \pm 0.00066)\%$. Both zero-field controls give zero focusing at the stored precision.

In Fig. 2, azimuthal pulls against GUINEA-PIG have RMS 1.39 and maximum absolute value 3.16. The distribution has RMS 1.35 but retains an endpoint pull of 9.63. A direct extraction of all 300 histogram bins from the source PDF confirms the sharp fall near 25 μrad: the largest edge discrepancy is $6.4 \times 10^{-5}\,\mu$rad, and the normalized-density difference is below $6.5 \times 10^{-8}\,\mu\text{rad}^{-1}$. The endpoint disagreement is not a digitization shift. These descriptive comparisons do not establish precision reproduction. Although the ensemble contains ten times the previous number of tracks, the simulated azimuth-bin SEM remains 7.9 times the estimated reference error at the median bin. More independent fields and probes, together with the input and grid studies in Appendix **??**, remain priorities.

A separate radiation-free test imposes the LEP annual mean $\langle \sigma_W^2 \rangle = 3044\,\text{MeV}^2$ [17] using independent Gaussian beams. The output width is 54.7956 MeV, versus $\sqrt{3044} = 55.1725$ MeV: a -0.683% difference with $D = 0.00385$. This checks propagation of an input width; it does not predict storage-ring equilibrium.

3.3 ILC pairs and the 1 TeV spectrum

The 500 GeV pair configuration uses the parameter set labelled USSC in Table I of Ref. [8]: 2×10^{10} particles per bunch, $\beta_{x,y} = (15, 0.4)$ mm and $\sigma_z = 300\,\mu$m. We compare the finite-mass EPA model and lowest-order matrix-element model of Sec. 1.2, with twelve independent collisions per model. The seed banks are disjoint. Common finite-beam and photon upgrades make this a controlled model comparison rather than an exact rerun of the historical implementation. Only incoherent daughters enter; other backgrounds and spectator beam leptons are excluded. The 5 MeV selection acts on each daughter at production, with no final-energy recut.

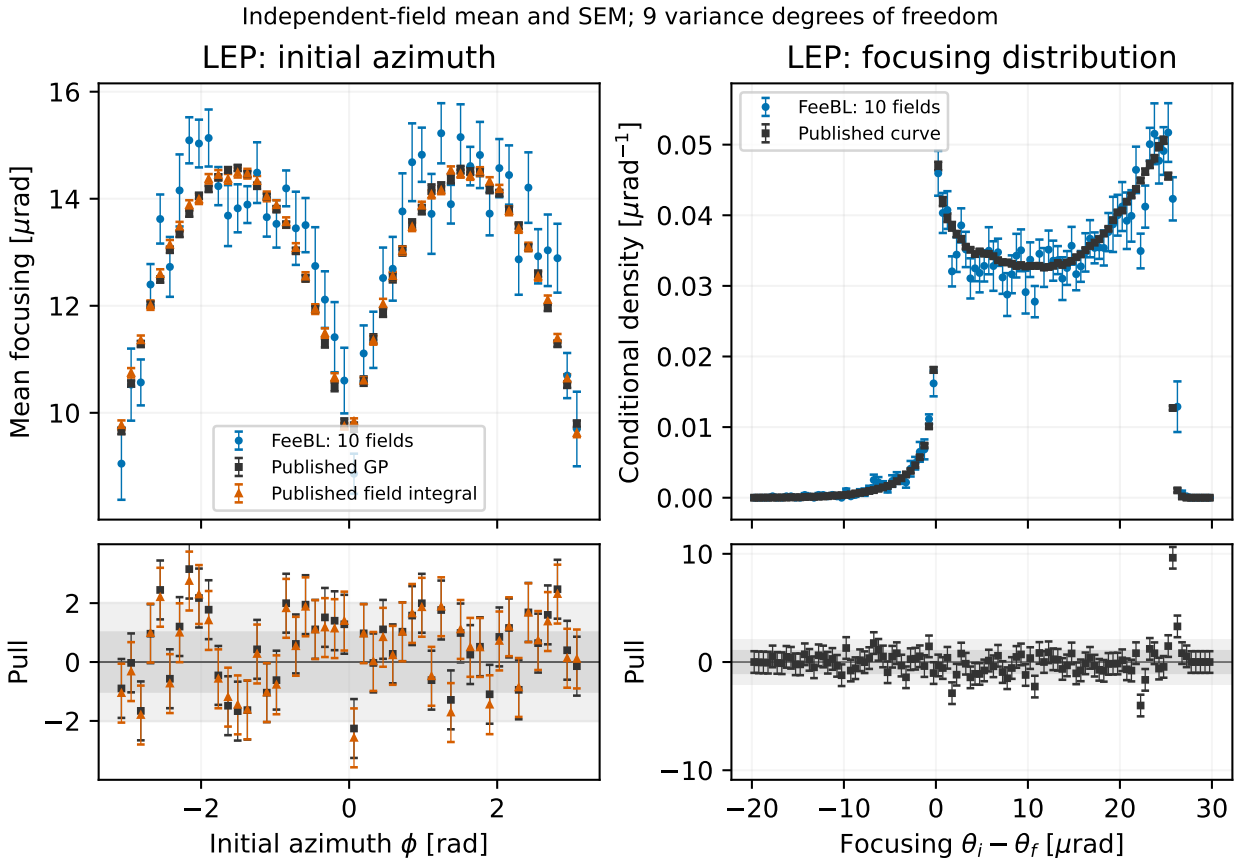


Figure 2: LEP electron focusing at an initial polar angle of 31.3 mrad. Left: mean focusing versus initial azimuth. Right: the conditional all-time distribution. The references are the published GUINEA-PIG and field-integration calculations [1]. FEEBL points average ten independent fields, preserving event and boundary correlations; their errors are standard errors of the mean. Reference errors combine local curve fluctuations with graphical-reading uncertainty. Lower panels include both contributions to Eq. (10) and have capped unit error bars.

The EPA calculation uses the published GUINEA-PIG prescription $Q^2_{\max} = s_{\text{pair}}/4$ with beam-size suppression, whereas the cited CAIN calculation uses $Q^2_{\max} = m_e^2$ [8]. Mode 2 does not apply the EPA hard virtuality veto. These model differences are separate from sampling error. Following Table II of Ref. [8], all three channels are compared as effective daughter-production cross sections, $\sigma_j^{\text{eff}} = \sum_a N_{j,a} / \sum_a \mathcal{L}_{ee,a}$. In particular, the published Breit–Wheeler and Bethe–Heitler entries do not use photon luminosities as their denominators. The yield/luminosity covariance is retained; a leave-one-collision-out jackknife checks the propagated error.

Table 7: Effective secondary-daughter production cross sections in mb for ILC at 500 GeV, normalized to electron–positron luminosity. Reference cross sections are the printed Table II values of Ref. [8]; totals sum the displayed channels. Each FEEBL model uses twelve independent collisions. Errors retain the complete channel and luminosity covariance. The final row reports luminosity per crossing separately.

Quantity	GUINEA-PIG	CAIN	FEEBL model 1	FEEBL model 2
Breit–Wheeler	1.05	1.04	1.163 ± 0.010	1.125 ± 0.019
Bethe–Heitler	37.70	34.50	38.870 ± 0.161	39.187 ± 0.272
Landau–Lifshitz	19.20	15.20	18.168 ± 0.048	18.839 ± 0.046
Total	57.95	50.74	58.201 ± 0.198	59.151 ± 0.276
$\mathcal{L}_{\text{BX}}$ [μb^{-1}]	1.82	1.82	1.7348 ± 0.0116	1.7443 ± 0.0095

Leave-one-out means and standard errors, half-sample comparisons and the largest kernel contributions test sensitivity to rare weights; all seeds are retained. Twelve seeds give eleven

variance degrees of freedom. These checks address sampling stability, not spatial or time convergence. Omitting one collision shifts each normalized ILC cross section by at most 0.70 of its full-sample SEM, while the estimated SEM changes by up to 21%. Finite-ensemble uncertainty in the uncertainty itself therefore remains material even without an individually dominant seed.

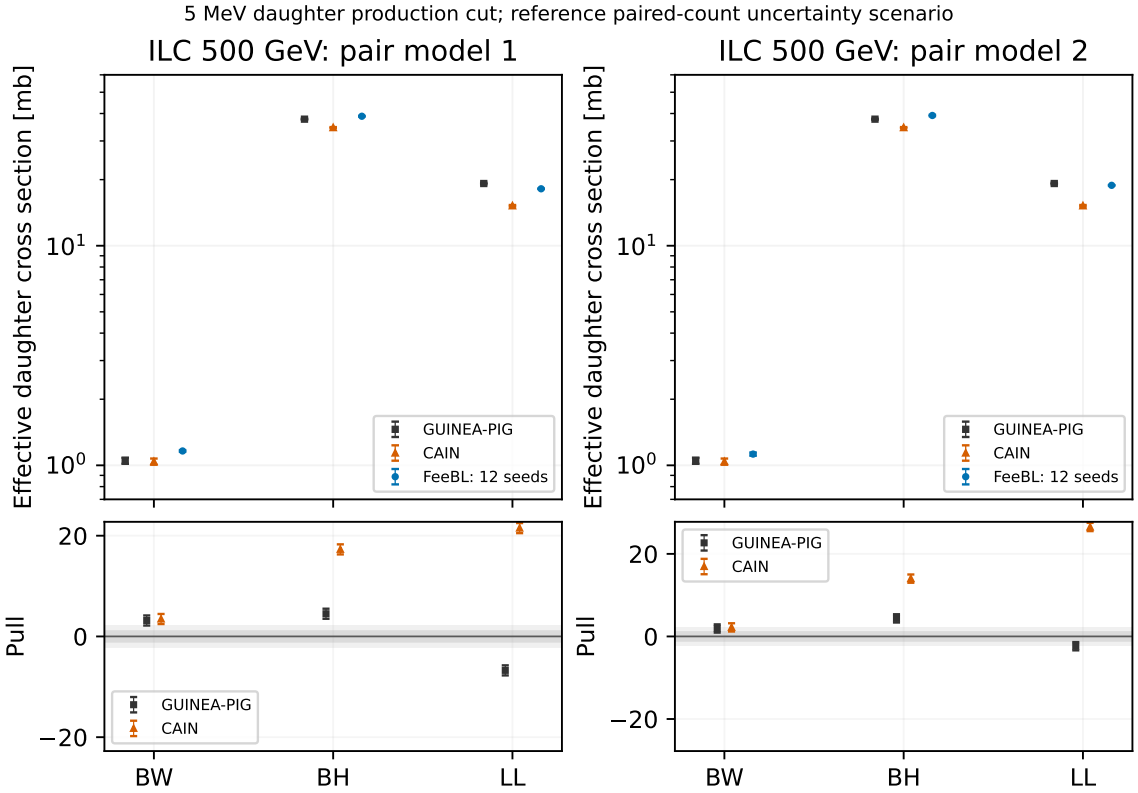


Figure 3: ILC effective daughter-production cross sections for the finite-mass EPA model (left) and lowest-order matrix-element model (right). Each FeeBL point divides the summed yield by summed electron–positron luminosity across twelve complete collisions, with the joint sampling error. The references use different virtual-photon prescriptions [8]. Pulls include the measured FeeBL covariance and an explicitly assumed reference count variance $2Y$, propagated through the fixed published luminosity; variations to Y and $4Y$ accompany the numerical results. These scenarios do not recover the original reference simulation's statistical errors. Lower bars are capped and show ±1.

The mode-2 Landau–Lifshitz cross section is 18.839 ± 0.046 mb, 1.88% below the published Guinea-Pig value of 19.2 mb. The corresponding comparison pull is -2.37 under the stated reference-error scenario. Its raw per-crossing yield is 6.11% lower, but the simulated luminosity is also 4.16% lower; comparing those counts alone conflates exposure and production physics. Their seed correlation is 0.895, so omitting the yield/luminosity covariance would overestimate the cross-section error by a factor of about three. The normalized mode-2 Breit–Wheeler and Bethe–Heitler results exceed Guinea-Pig by 7.15% and 3.94%, respectively.

The reference inter-code spread also has an identified physical source. With the Cain virtuality limit, Ref. [8] obtains an LL cross section of 15.6 mb in Guinea-Pig, close to Cain's 15.2 mb, whereas the Guinea-Pig default gives 19.2 mb. That spread is therefore not a statistical error bar. The CAIN manual describes an almost-real-photon approximation with an electron-mass transverse-momentum scale and a nonlocal transverse photon distribution involving K_1^2 [18, Secs. 5.11.2–5.11.3]. Changing the virtuality limit alone does not reproduce that spatial model; the finite-beam prescription remains a separate comparison variable. The following controls separate these effects in the present implementation.

Controlled pair-process comparison. We further separate the implementation choices with nine configurations, each using four matched collision seeds. All 36 calculations retain the 5 MeV daughter-production threshold and use unpolarized real photons; the latter permits the legacy EPA control. Figure 4 shows all three channels normalized to their measured electron–positron luminosity. Birth and final records close separately by process and species. The shared-seed covariance enters each difference; changing a model also changes subsequent random-number consumption, so exact cancellation is neither assumed nor required.

For LL, the refined EPA treatment changes the ring-placement result by -1.321 ± 0.089 mb, or -6.66%. This combines the finite-mass flux, angular-sampling and coupling-constant changes. Replacing ring placement with the impact-parameter profile changes EPA by -0.450 ± 0.081 mb. The full hard-process treatment then changes the profile result by $+0.852 \pm 0.046$ mb. These shifts depend on the starting model: the difference between the EPA and full-amplitude profile changes is 0.283 ± 0.081 mb. The packaged paired-change table retains this interaction and the complete covariance.

The full-amplitude ring result is 19.087 ± 0.103 mb, 0.59% below the published Guinea-Pig value. With beam-size suppression disabled, it becomes 31.502 ± 0.108 mb, 0.94% below the independent BDK calculation's 31.8 mb in Ref. [8]. BDK includes electroweak diagrams beyond the present QED scope. These comparisons support the normalization and overall scale, without supplying missing reference errors or proving collider convergence. The exact calculation also uses a minimum-virtuality spectator assignment to define the produced daughters in identical-lepton final states; this selection is distinct from the EPA construction.

The electron-mass EPA cutoff lowers LL by 4.063 ± 0.093 mb. This large prescription dependence is consistent with the source's cutoff study, although the refined EPA and CAIN spatial models remain different. Doubling the transverse domain and cell counts at fixed spacing changes the full-profile LL result by -0.123 ± 0.081 mb, or -0.65%. That control includes both field/source coverage and virtual-photon acceptance. No additional LL kernel defect is identified by these checks. The remaining percent-scale difference is compatible with the measured model and domain sensitivities; finer grids, more seeds and matched source inputs are still required for a precision comparison.

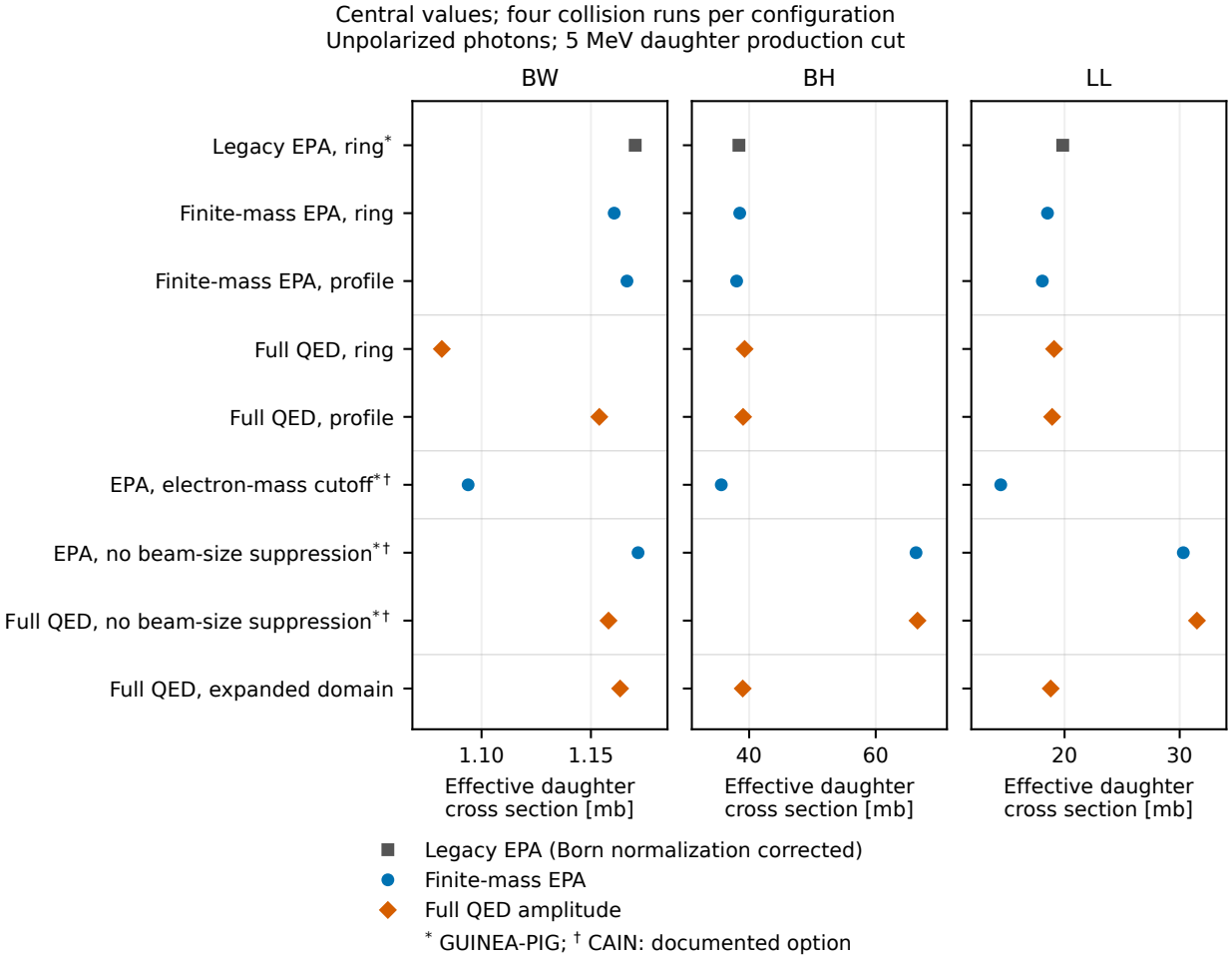


Figure 4: Central effective daughter-production cross sections for nine ILC 500 GeV configurations, each using four matched collision runs. Each point divides the full summed daughter yield by the summed electron–positron luminosity. Statistical bars are omitted; sampling SEMs and paired covariance remain in the accompanying numerical tables. Superscripts * and † identify a corresponding prescription or the marked option documented for GUINEA-PIG and CAIN, respectively [8, 18], rather than equivalent complete configurations. Unmarked rows do not by themselves establish novelty: CAIN already includes nonlocal virtual-photon sampling, and domain enlargement is a numerical control. The refined EPA and full hard-process treatments extend the inherited implementation. The legacy EPA control already contains the corrected Born normalization. "Full QED" refers to hard amplitudes; finite-beam placement remains a separate model. Disabling beam-size suppression places photons locally; it does not make the bunches infinite. The expanded domain retains cell spacing. Real photons are unpolarized, unlike Fig. 3. The BW hard kernel is unchanged between modes 1 and 2; variations include sampling of the beam and photon histories.

At 1 TeV, the Gaussian reconstruction of the ILC Technical Design Report's B1b upgrade parameter set gives full/peak spectrum distances $D = 0.0259/0.0182$ relative to the Forward Calorimetry (FCAL) collaboration's distribution [19]. The corresponding conditional first Wasserstein distances are 2.51/0.160 GeV for the mean of 12 independent collisions. Broad tail and peak shapes are similar, but the non-Gaussian reference input and its correlations are not reconstructed. More macroparticles alone cannot repair that input mismatch. At the retained fine comparison binning, the full-spectrum pull RMS is 1.82, with a maximum of 18.3 near the endpoint and magnitudes above 10 in adjacent bins. The peak-panel maximum is 3.10. These are diagnostic differences with estimated reference errors, not source-certified significances or evidence of precision agreement.

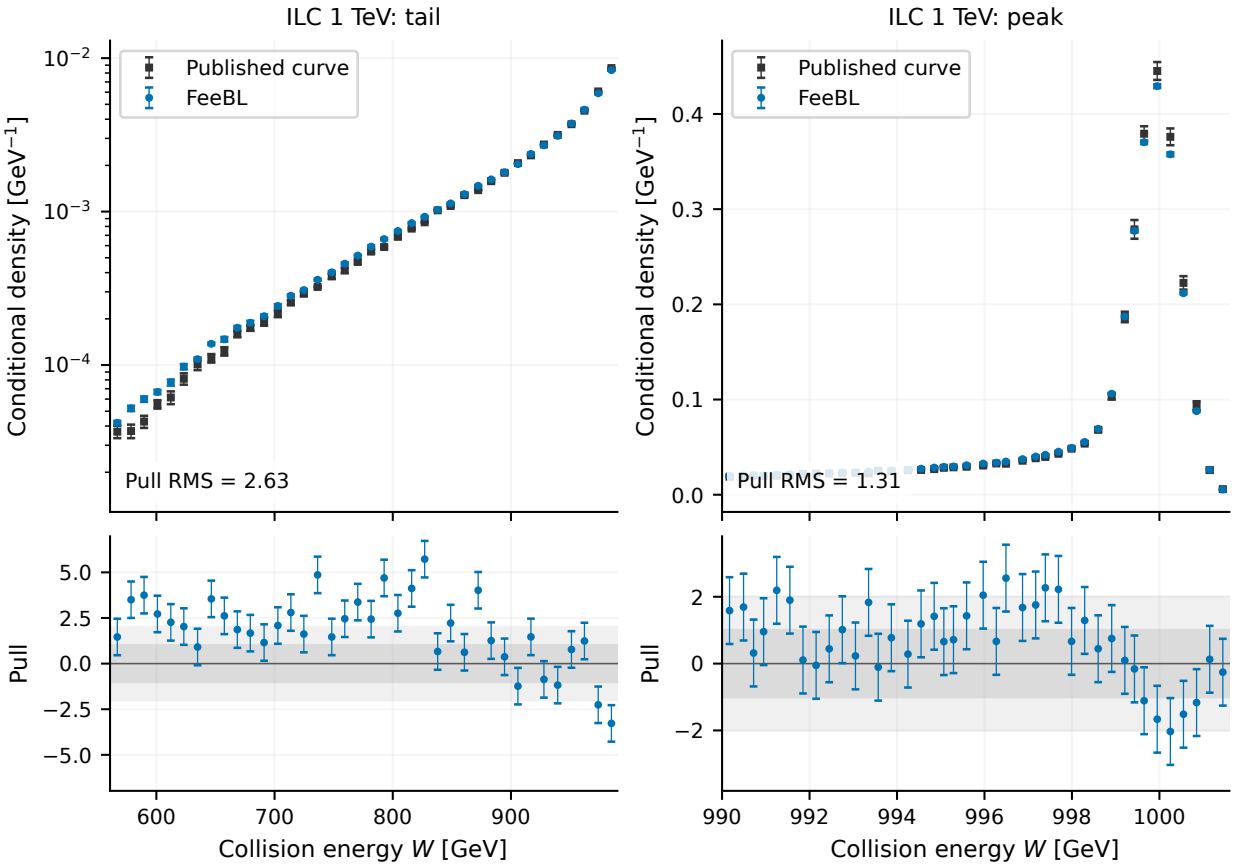


Figure 5: Tail and peak ILC 1 TeV luminosity spectra compared with Ref. [19]. Panels retain normalization on their original reference intervals; censored intervals remain gaps. Upper errors and lower pulls use the complete-collision sampling covariance and local reference-fluctuation estimates. The left panel shows 38 tail bins from 560.8 to 990 GeV and excludes the peak region shown in the right panel, which uses 40 bins from 990 to 1001.6 GeV. The display cuts retain the original panel normalization and enter the bin-overlap covariance transfer before recomputing pulls; the uncut numerical comparison is preserved. Every pull includes both variances and has a capped unit error bar.

The two digitized reference panels are also checked against one another. On complete common coarse bins from 981.24 to 1000.65 GeV, their conditional CDF distance is 0.0374, compared with 0.00083 when the same native records are remapped through the two binnings. This interval retains 93.7% of the peak-panel weight and excludes the final flagged reference segment. The inconsistency therefore extends beyond one endpoint bin. Fractional edge-bin clipping increases the reference distance to 0.0664, illustrating an additional rebinning assumption. Both audits are retained; neither justifies changing a published ordinate to improve agreement.

3.4 HALHF luminosity and particle populations

The two panels of Fig. 5 resolve the broad tail and high-energy peak independently; their conditional normalizations differ. The September 2026 HALHF parameters [20] provide a stronger-field, asymmetric-energy comparison. Gaussian incoming beams reproduce their moments, not the full imported distributions. Beamstrahlung, ISR, incoherent/coherent pairs and Compton processes follow the declared cards. High-energy fractions in Table 8 use incoming energies before optional ISR.

Table 8: Historical HALHF luminosity in $10^{34}\,\mathrm{cm}^{-2}\mathrm{s}^{-1}$, pre-ISR fraction $f_{0.99}$ and conditional spectral distances. Each result has one seed. The 250 GeV row uses the expanded transverse domain.

W_0 [GeV]	$\mathcal{L}_{\mathrm{ref}}$	$\mathcal{L}$	Difference	$f_{0.99,\mathrm{ref}}$	$f_{0.99}$	D_{full}	D_{peak}
250	1.14	1.11208	−2.45%	0.66	0.65212	0.00940	0.01442
380	1.75	1.71061	−2.25%	0.57	0.55784	0.01151	0.01599
550	2.55	2.46365	−3.39%	0.50	0.48812	0.01281	0.01223

The agreement is plausible at the few-percent luminosity level. Applying the energy selection after ISR instead gives fractions 0.43649, 0.36573 and 0.31699, demonstrating the importance of matching radiation definitions. At 250 GeV, expanding the horizontal/vertical domains by factors 2/8 at fixed cell spacing changes luminosity by +0.469% and $f_{0.99}$ by −1.435%. A separate compact-domain particle/slice/time refinement changes them by +2.599% and −2.094%. These do not establish convergence. At 380 and 550 GeV, maximum outside-main-mesh fractions in individual slice encounters are 30.8% and 37.5%. Such maxima are not fractions of total lost bunch charge, but require explicit domain tests.

Figure 6 displays the broad shape agreement at 250 GeV; this does not resolve the separate few-percent luminosity shifts. Figure 7 isolates incoherent daughters in both pair models, using twelve independent collisions per model. Final-position selections $z > 0$ and $z < 0$ follow the reference panel labels. We reconstruct the angle toward the nearer beam axis, $\theta = \mathrm{atan2}(p_T, |p_z|)$, with $p_T = E\sqrt{\beta_x^2 + \beta_y^2}$ for positive total energy E. The reference does not fully specify its folding algorithm, so this remains a declared reconstruction of its convention.

The stored fraction is 0.1; per-crossing count estimates include its inverse weight, without a final-energy recut. Reference image cells indicate occupied support rather than event counts or a normalized density. The simulated panels therefore use the fraction of independent stored samples occupying each cell. Larger union support alone is not evidence of improved agreement. Position/momentum-sign differences, logarithmic exclusions and flow counts accompany the numerical output. At the displayed cell size, 97.8–99.2% of the in-range simulated counts fall in published occupied cells. The union of twelve samples covers 73.8–86.6% of the reference cells across species, hemispheres and models. The dominant bands are plausible, but these support fractions depend on sample size and do not measure density agreement. Position and momentum signs differ for about 1.25% of stored records; the position selection is retained. Field-domain convergence and rare-cell sampling remain open. Published HALHF $\chi(t)$ uses instantaneous whole-bunch quantities, whereas the present histograms sample slice-encounter field kicks. A quantitative time-dependent comparison requires synchronous definitions and common incoming records.

3.5 Kinematic and measurement-level controls

Independent Gaussian relative energy spreads of 0.19% and 0.152% with a 14 mrad full crossing angle give $\sigma_W = 0.304141\,\mathrm{GeV}$ at 250 GeV, or 0.121657%, by Gaussian quadrature. This reproduces the rounded 0.122% input-width expectation in Ref. [21]. A 512-event test of massive $\mu^+\mu^-\gamma$ reconstruction identities from that reference closes to a maximum $|\Delta W| = 2.84 \times 10^{-12}\,\mathrm{GeV}$. These are input and algebra checks, not a reproduction of its fitted process-weighted peak width or

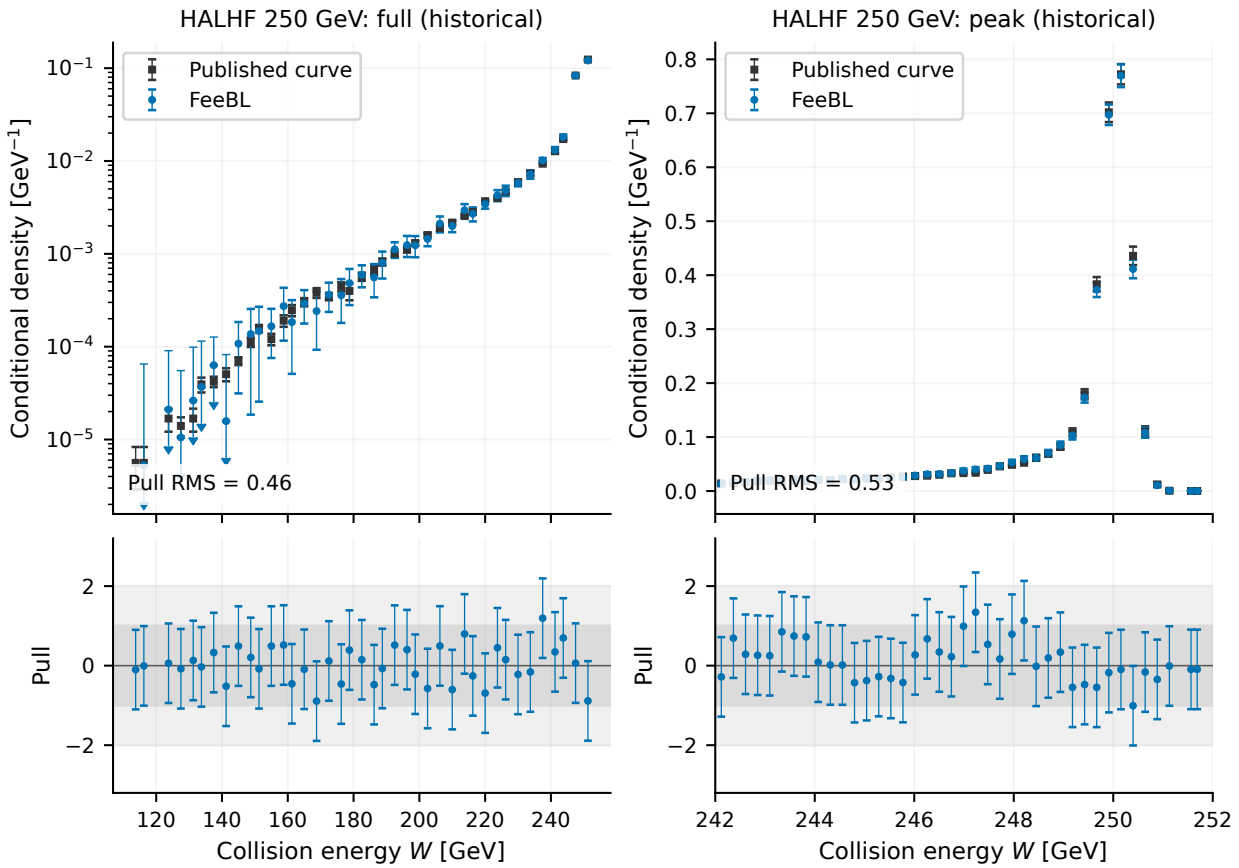


Figure 6: HALHF 250 GeV full and peak spectra relative to Ref. [20], using the expanded domain. Conditional normalization, estimated marginal errors and unit-bar pulls follow the ILC comparison. Both panels use 40 display bins, with a 242–252 GeV peak axis. The source and simulated incoming moments agree, while unprovided particle-level correlations remain a qualification.

detector precision.

The ILC orbital-deflection study in Sec. 4.3.1.1 of Ref. [16] provides a distinct transport benchmark. It uses two million radiative Bhabha events generated with BHLUMI over 1–6 degrees. At $W_0 = 250\,\mathrm{GeV}$, its inner-boundary median changes in global polar angle are $-96.1 \pm 0.969\,\mu\mathrm{rad}$ for electrons and $+96.1 \pm 1.03\,\mu\mathrm{rad}$ for positrons (Fig. 4.7). Here the change is $\theta_f - \theta_i$, with both angles measured from the same global $+z$ axis; the opposite signs describe focusing in the two arms. The displayed errors characterize the source medians, while its repeated-run scatter is approximately 3%. The packaged monoenergetic, back-to-back probe example tests transport and visualization; it does not reproduce this radiative population. A matched comparison requires the original bunch population, waist settings and boundary selections in addition to the published energy and beam-size parameters. The extracted reference curves are supplied for that comparison, which remains to be completed.

The eventwise deflection correction in Ref. [22] additionally uses radiative Bhabha events, detector response and a trained regression. Its quoted measurement precision is not inferred from the present beam-only tests. Reproduction requires the event selection, response model and training/validation samples as well as a collision calculation.

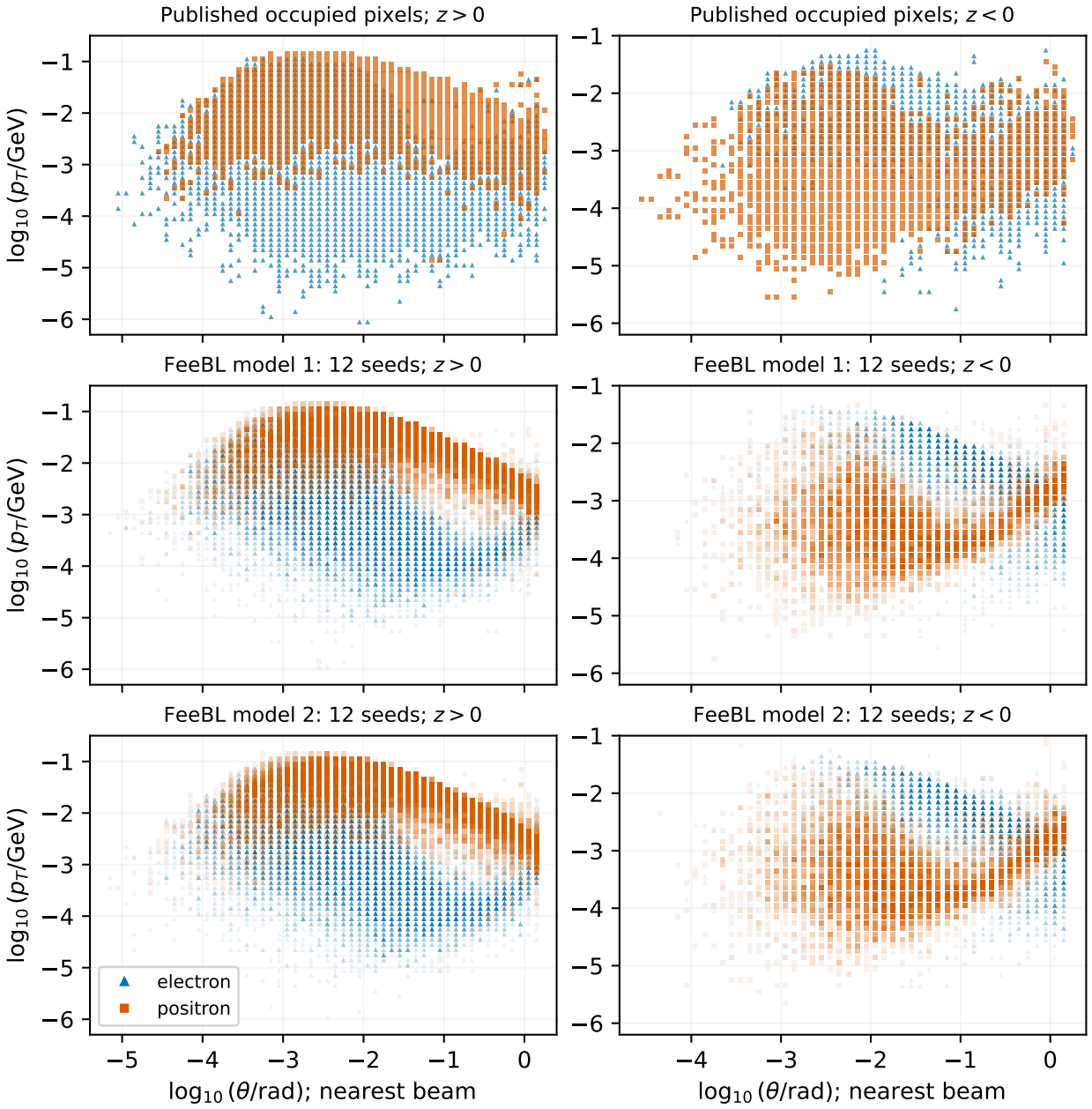


Figure 7: Polar-angle and transverse-momentum support of incoherent pairs for HALHF at 250 GeV. The top row shows the published occupied cells [20]; the middle and bottom rows show FeeBL pair modes 1 and 2. Columns select final position $z > 0$ and $z < 0$; the angle is folded toward the nearer beam axis. Color and marker shape distinguish electrons and positrons. In simulated panels, opacity is the fraction of twelve independent stored samples occupying a cell. The comparison concerns graphical support; reference pixels do not supply count uncertainties or a normalized density.

3.6 CPU and GPU comparison

We compare eight CPU and GPU runs with different seeds using FCC-ee inputs from Ref. [23]: 120 GeV per beam, 30 mrad full crossing angle and 50% crab-waist strength. The FCC-ee inputs are particularly useful for a stress and consistency test as the long bunch length and narrow transverse RMS spread, coupled with the crab-waist scheme, resulted in FCC-ee simulations being slowest and greediest in terms of resources needed to achieve numerical convergence. We use the published equilibrium RMS moments for the beams; ring equilibrium is not calculated nor currently supported. Each beam starts with 10^6 macroparticles and cut at three RMS bunch lengths. The mesh has 512×1024 transverse cells, 64 slices and two substeps. We disable spin and beam pairs for these runs and examine only the luminosity spectra.

For smaller scale local tests, such as most of the tests in this paper, run time was observed as 2.5 ± 1.1 minutes, while being run on an average home computer. The stress testing done here was done on DESY Maxwell using two Intel Xeon Silver 4216 processors and two 32 GiB NVIDIA V100 SXM2 GPUs. Two independent collisions run concurrently, each on 16 physical cores and, for GPU execution, one GPU. Both tests use the same executable, built with GCC 12.2 and CUDA 12.9. The full allocation, including build and trials, was 32 minutes and 28 seconds.

Table 9 presents the timing results. The median time ratio is 1.82 with a range of 1.79 to 1.86. Host physics still occupies 44.45 seconds per GPU assisted collision and thus approximately 68% of elapsed time. This results means that further optimizations are needed before FeeBL is limited by GPU saturation and, therefore, we expect future development will improve this performance further.

Table 9: FCC-ee stress test timing results. Collision time is the mean from eight seeds; other rows sum eight collisions or four concurrent two collision batches. CPU consumption is user plus system time. Assigned GPU time includes host work and idle periods. CUDA intervals include cuFFT but are not total collision times. We round to the nearest second for presentation clarity.

Time on stage and on device [s]	CPU	GPU with host threads
Mean elapsed collision time	118.46 ± 0.67	65.106 ± 0.069
Elapsed time per two collision batch	119.39 ± 0.66	65.65 ± 0.29
CPU core time per collision	1045.1 ± 8.5	237.46 ± 0.27
CUDA kernel and cuFFT event intervals per collision	—	6.42634 ± 0.00032
Host and device transfer event intervals per collision	—	0.79595 ± 0.00086
GPU device time per collision	—	65.106 ± 0.069

Across all eight seeds the mean luminosity per crossing is $(9.1143 \pm 0.0012) \times 10^{32}\,\mathrm{m}^{-2}$. Each matched histogram is identical at six-significant-digit output precision; this paired closure is retained separately from Fig. 8. That figure uses four odd-seed CPU runs and four even-seed GPU runs, with no shared seeds. Its difference covariance is the sum of the two covariance matrices of the mean, without a paired cross term. The displayed subtraction is CPU minus GPU, with largest absolute pull 1.66. The timing table retains all eight runs per backend. When GPU is enabled in the input card the resulting FeeBL simulation currently uses only one GPU; multi-GPU execution is planned for a future release.

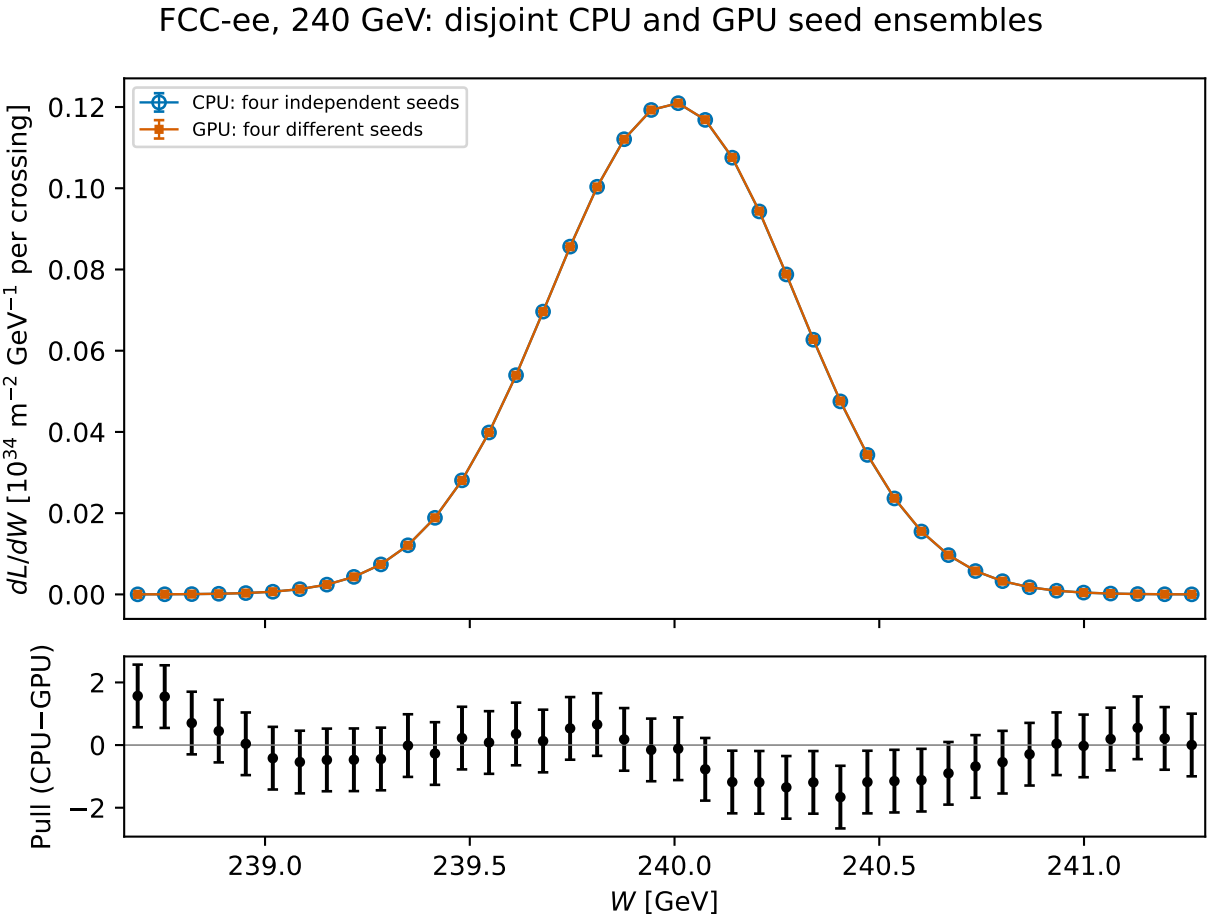


Figure 8: FCC-ee ZH comparison of independent ensembles of four runs, using one million macroparticles per beam and a $512 \times 1024 \times 64$ cell mesh. The displayed energy interval has forty 66 MeV bins. Blue open circles show the 16-thread CPU result and orange squares show the result from one V100 with 16 host threads. The CPU seeds are 72942001, 003, 005 and 007 while the GPU seeds are 72942002, 004, 006 and 008. Errors are standard errors across complete collisions. Pulls are CPU minus GPU divided by the combined standard error.

4 Depolarization and independent spin tests

The ILC comparison requires matching the beam parameters, spin observable and energy selection before interpreting a difference as incorrect spin physics. The 2006 calculation reports a luminosity-weighted loss of 0.10%, comprising 0.08% from precession and 0.02% from spin flips [24]. It does not specify an energy cut or an explicit per-beam versus combined formula. The later implementation paper defines $1 - \sqrt{\langle P_{z1} P_{z2} \rangle}$ and gives approximately 0.23% for GUINEA-PIG++ and 0.24% for CAIN [5]. These are different historical comparisons, not interchangeable estimates to be pooled without their conditions. The earlier table also gives 0.19% for Large Y and 0.24% for Low P; the later code values are 0.46–0.57% and 0.41–0.50%, respectively. The repeated change of scale motivates examining historical definitions and model settings rather than assigning the nominal difference to one implementation defect. Matching scenario names does not establish identical optics. In particular, equivalence of the early Low Q and later Low N sets has not been established.

Table 10: Published nominal ILC predictions for initially fully polarized beams. Values are absolute losses expressed in %. Only the later spin-product definition is explicit. The 2006 table does not provide an uncertainty.

Publication/code	β_x [mm]	Precession [%]	Total [%]
EPAC 2006, CAIN [24]	21	0.08	0.10
Rimbault et al. 2009, GUINEA-PIG++ [5]	20	0.17	0.23
Rimbault et al. 2009, CAIN [5]	20	0.19	0.24
PAC 2011, CAIN [25]	20	0.17	0.22

The 2009 conclusion quotes $0.23 \pm 0.01\%$ without defining a confidence level. The 2011 study estimates an absolute statistical uncertainty near 0.10%. The EPAC 2008 result $0.21 \pm 0.01 \pm 0.03\%$ instead expresses model sensitivities to field and spin-flip energy-ratio variations [26]; these cannot supply an uncertainty for the 2006 value.

For initially longitudinal, flat, head-on beams with weak horizontal disruption, Yokoya and Chen obtain

$$\Delta P_{\mathcal{L}} = \frac{2}{3}\frac{\ln(9/8)}{\ln(4/3)} \Delta P_{\text{out}} \simeq 0.273\, \Delta P_{\text{out}}, \tag{14}$$

where ΔP_{out} is the final particle-average loss of one beam [27, Eq. (16)]. This inclusive, small-loss relation is not an effective-polarization or energy-cut conversion. It need not hold when both disruption parameters are large, and its extension to spin flips is approximate. The 2006 proceedings print the two labels in the reverse order relative to this derivation, although their Table 2 is explicitly labelled luminosity weighted [24]. That inconsistency does not justify rescaling the tabulated results or establish the cause of the discrepancy.

The 125 GeV-per-beam table in Yokoya's presentation [28] separately quotes electron and positron depolarization, but its weighting, spin projection and normalization must be resolved before a numerical comparison with the collision-weighted loss here. A published precision of two decimal places is not a statistical uncertainty.

4.1 Spin equation and corrections tested independently

Classical precession follows the Thomas–Bargmann–Michel–Telegdi equation [29]. In SI units, one convention is

$$\frac{d\boldsymbol{s}}{dt} = \boldsymbol{\Omega} \times \boldsymbol{s}, \tag{15}$$

$$\boldsymbol{\Omega} = -\frac{q}{m}\left[\left(a + \frac{1}{\gamma}\right)\boldsymbol{B} - \frac{a\gamma}{\gamma+1}(\boldsymbol{\beta}\cdot\boldsymbol{B})\boldsymbol{\beta} - \left(a + \frac{1}{\gamma+1}\right)\frac{\boldsymbol{\beta}\times\boldsymbol{E}}{c}\right], \tag{16}$$

where q is signed and $a = (g-2)/2$. Beamstrahlung spin transitions and conditional no-emission evolution follow the stated CAIN radiation model [30, 18]. The geometry, basis and recoil update are therefore tested independently of a collider's final scalar loss.

Three implementation defects were corrected before the results reported here. A common unit ultrarelativistic direction now defines both spin field projections and the radiation basis, avoiding inconsistent use of $(v_x/c, v_y/c, \pm 1)$. The first transverse radiation basis vector follows the acceleration applied by the orbit update, with $\boldsymbol{e}_2 = \widehat{\boldsymbol{p}} \times \boldsymbol{e}_1$. Finally, multiple emissions within an interval update the quantum parameter after recoil, $\chi = \chi_{\mathrm{in}} E/E_{\mathrm{in}}$ at fixed field and direction. The energy was previously updated while χ retained its incident value. This correction applies to spin-dependent and spin-independent radiation; it does not tune the single-emission rate to a collider benchmark.

Independent adaptive Bessel-function quadrature tests the transition law. The quantum parameters are $\chi = 0.01, 0.05, 0.2, 1, 3$, with two proposal exposures and four initial spin ensembles. Forty settings each contain two million trials, giving 200 probability, radiated-energy and spin means plus 40 norm/energy checks. A retrospective familywise assessment uses a fixed 5% simultaneous empirical–Bernstein criterion [31]. All 200 means satisfy the criterion; the largest absolute discrepancy is 0.706 times its simultaneous sampling radius. Finite-step and numerical allowances remain separate, and all 40 bounds checks report zero invalid trials. This conservative result assumes independent bounded trials within each setting; it does not require independence between settings or establish the absence of smaller biases. The retained records and complete calculation are in `benchmarks/analysis/HISTORICAL_SAMPLER.md`. Twenty-three geometry/transport checks pass after correction; fourteen failed before it. An independent oblique-field BMT calculation agrees to 1.1×10^{-16} in spin components. Four weak-field checks recover longitudinal/transverse damping 0.777716 versus 7/9, and equilibrium polarization 0.923734 versus $8/(5\sqrt{3})$. Five anomalous-moment checks test the implemented approximation and asymptotes [32].

Figure 9 uses an independent-seed replication rather than reusing one seed across settings. Its photon-probability and radiated-energy pull RMS values are 1.0053 and 1.0012. For initial $S_y = +1$, the mean pulls across 40 independent runs are -0.116 ± 0.169 for photon probability and -0.232 ± 0.141 for radiated energy. Here the errors are empirical SEMs across the fixed quantum-parameter and exposure settings. The latter is the largest subset offset, at 1.65 SEMs. These results do not resolve a common bias in that subset, although smaller or setting-dependent biases remain possible. S_y is the transverse radiation-basis spin component in this fixture, not longitudinal beam helicity. No offset was subtracted or error rescaled. The portable producer recomputes the quadrature from the stored sufficient statistics. In the original quadrature-conditioning test, changes were below 1.28×10^{-10} sampling-error units. The controlled radiation test therefore has substantially better understood sampling errors than a digitized collider curve, but does not establish all strong-field QED or low-energy transport effects.

An extended fixed-field test at $E_0 = 250\,\mathrm{GeV}$, $\chi_{\mathrm{in}} = 1$ uses 200 000 particles per setting. Grouping the same exposure in one or 64 external intervals gives mean energy fractions 0.503881/0.528189 before the recoil correction and 0.527871/0.528447 after it. Longitudinal spin means change from 0.866271/0.884942 to 0.884420/0.885002. This deliberately large exposure shows why recoil consistency matters, even when a particular ILC loss changes only modestly.

4.2 Inclusive and high-energy collision samples

At the original 10 000-particle settings, corrected per-beam helicity losses are 0.21350 and 0.22365%. The fixed-axis spin-product loss is 0.21907%, versus 0.21846% for local helicity. Replacing the correlated fixed-axis product with separate mean projections changes the loss by only 0.000113% in this case. Final outgoing helicity losses are 0.7354/0.7662%, showing why a whole outgoing bunch cannot replace a collision-weighted average.

The energy selections in Table 11 use the instantaneous massive invariant energy, including

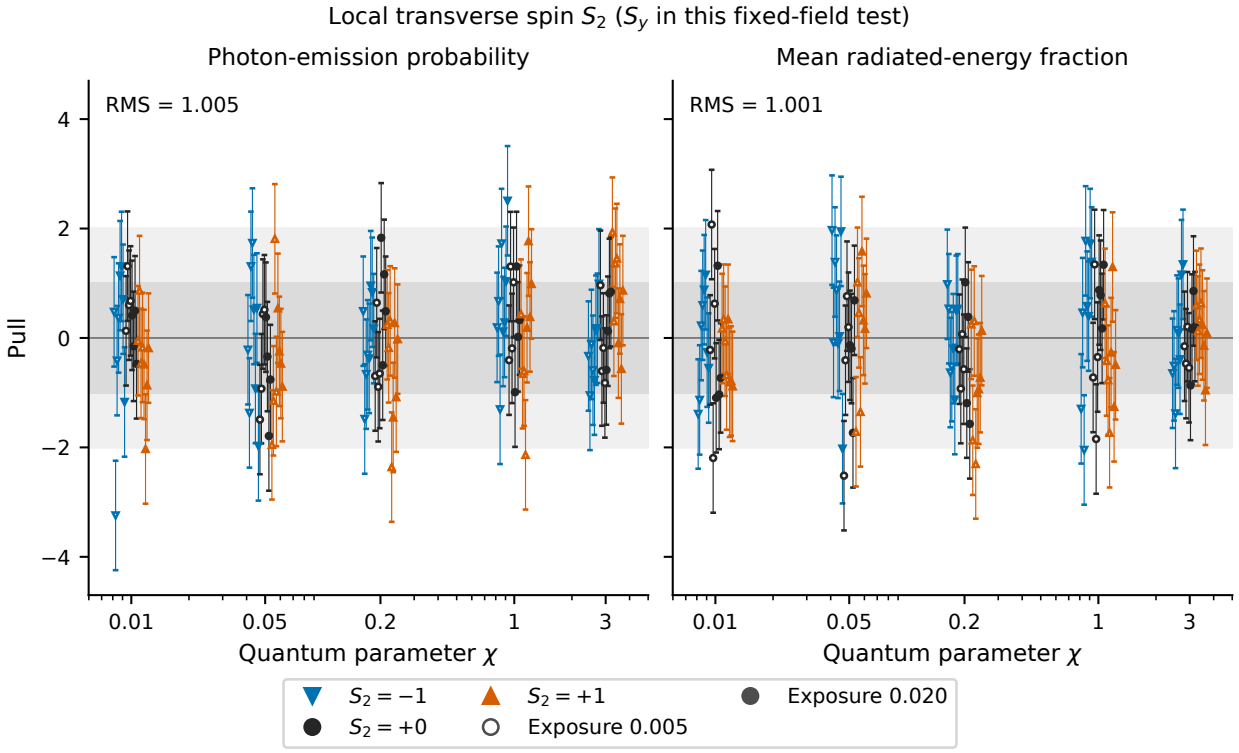


Figure 9: Sampling pulls for photon probability and mean radiated-energy fraction relative to independent deterministic quadrature. There are 120 independent runs: three initial transverse spin projections, five quantum parameters, two exposures and four fresh seeds per setting, each with two million trials. Both uncertainty contributions enter the denominator; the quadrature target has zero sampling variance and separately checked numerical accuracy. Capped bars are ±1. Probability and radiated-energy estimates share trials within a run; seeds are independent across runs.

beamstrahlung and excluding ISR, at $W_0 = 500\,\mathrm{GeV}$. They retain no additional single-beam cut. Above 98% of nominal energy, the mean local spin-product loss is 31% smaller than its inclusive value; above 99%, it is 35% smaller. Neither selection produces the historical 0.10% value. The tested cut dependence is substantial, but cannot identify an unreported historical acceptance.

Table 11: Three-seed energy-selected results with 40 000 macroparticles per beam at fixed coarse mesh resolution. Losses are in % relative to unit helicity. Errors are SEMs of the seed mean only. The product uses local helicities, not conventional effective polarization.

Selection	Retained $\mathcal{L}$ [%]	ΔP_-	ΔP_+	ΔP_{pair}
Inclusive	100	0.2228 ± 0.0018	0.2218 ± 0.0037	0.2222 ± 0.0018
$W \geq 0.98W_0$	83.40 ± 0.11	0.1534 ± 0.0015	0.1536 ± 0.0022	0.1534 ± 0.0009
$W \geq 0.99W_0$	74.19 ± 0.10	0.1442 ± 0.0016	0.1443 ± 0.0021	0.1442 ± 0.0009

Directly accumulated per-beam moments provide the inclusive and 98% values. These statistical errors do not include the observed 7.15% inclusive-loss change under joint numerical refinement; no corresponding calibrated cut-specific numerical error has been established. Cut-dependent local products and the 99% per-beam values are reconstructed from 0.05 GeV bins aligned with the cuts; the additional rounding bound on the product loss is 2.54×10^{-5}%. Fixed-axis products were not retained as a function of energy, so the cut rows are explicitly local-helicity observables.

4.3 Numerical dependence and later optical configurations

Three independent 40 000-particle seeds give the inclusive fixed-axis loss $0.22282 \pm 0.00183\%$, with seed standard deviation 0.00318%. The error is the standard error of the mean (SEM), not a 95% interval. For three independent normal seed estimates, a 95% Student-t interval uses $t_{0.975,2} = 4.303$, giving approximately $[0.21495, 0.23069]\%$. The observed matched numerical refinement of +7.15% is a separate sensitivity, not a confidence interval or a calibrated total numerical error. Disabling spin flips gives $0.16752 \pm 0.00105\%$ and the paired difference $0.05530 \pm 0.00079\%$. This is a model-switch difference, not an exact additive decomposition: altered random consumption can change trajectories.

Table 12: Matched-seed numerical variations at $\beta_x = 21$ mm. N_m is macroparticles per beam, N_z slices and N_t encounter subdivisions. Shifts between rows are numerical sensitivities, not independent errors.

Variation	N_m	$N_x = N_y$	N_z	N_t	$\Delta P_{z,L}$ [%]
Baseline	10 000	128	32	2	0.21907
Particles	20 000	128	32	2	0.22245
Particles	40 000	128	32	2	0.21948
Transverse cells	40 000	256	32	2	0.22571
Slices	40 000	128	64	2	0.22457
Time subdivisions	40 000	128	32	4	0.22476
Combined	40 000	256	64	4	0.23518

The combined refinement increases the matched loss by 7.15%, exceeding the seed SEM. Numerical convergence is therefore unresolved even though the transport controls pass. Changing only β_x from 21 to 20 mm gives 0.22985%, a 4.72% increase, consistent with the 4.85% sensitivity of the approximate Yokoya–Chen precession estimate [27]. This small optical change does not explain the factor of two relative to 2006. The confirmed defects, axis conventions and tested cuts do not establish the origin of that historical discrepancy.

Table 13 specifies the additional 2009 optical configurations. Their evaluated transverse sizes use the published emittances and beta functions rather than rounded size entries.

Table 13: Inputs for the 2009 spin comparisons [5, Table 1]. Each beam has energy 250 GeV, zero intrinsic energy spread and unit initial longitudinal polarization. The crossing is head-on. Emittances are in mm mrad, beta functions in mm, transverse sizes in nm and bunch length in μm. All use 128^2 transverse cells, $N_z = 32$ and $N_t = 2$. The Low N sample has fewer macroparticles and requires separate refinement.

Case	$N/10^{10}$	$\epsilon_{nx}/\epsilon_{ny}$	β_x/β_y	σ_x/σ_y	σ_z	N_m
Nominal	2	10/0.04	20/0.4	639.4/5.72	300	40,000
Low N	1	10/0.03	11/0.2	474.2/3.50	200	10,000
Large Y	2	10/0.08	11/0.6	474.2/9.91	500	40,000
Low P	2	10/0.036	11/0.2	474.2/3.84	200	40,000

These additional configurations provide a more informative matched comparison. Their reported beam emittances and beta functions set the sizes; reconstructed Gaussian beams, zero energy spread and IP waists are used where the original particle records are absent. Large Y and Low P lie between the published code values and are assessed as plausible. Their 10 000-particle losses, 0.49964 and 0.51113 %, change by +1.54% and −4.19% when particle count is increased, restoring the published ordering. Nominal rounds to 0.23 % at the reference precision. Low N is 6.18% below both rounded references: it is plausible in scale, but is not within their zero printed spread. Including FeeBL in the three-code SD and then testing its own agreement against that SD would be circular. Shared physics and code heritage also prevent treating the three codes as independent measurements.

Figure 10 displays the per-beam selection dependence and the optical-configuration comparison

Table 14: Inclusive fixed-axis spin-product losses in % for the 2009 configurations [5]. Parentheses give the primary macroparticle count per beam. Published inter-code differences are descriptive; they are not uncertainty estimates or an equivalence test.

Optics	GUINEA-PIG++ [%]	CAIN [%]	FeeBL [%]	N_m
nominal	0.23	0.24	0.2298	40000
low n	0.13	0.13	0.1220	10000
large y	0.46	0.57	0.5073	40000
low p	0.41	0.50	0.4897	40000

separately. The available independent-seed per-beam results cover ILC. Extending this comparison to HALHF requires dedicated spin-enabled ensembles. The proposed comparison is restricted to ILC/LCF and HALHF. The current code rejects spin tracking with the canonical crab-waist configuration, so FCC-ee depolarization is not included. Support for polarized beams with crab-waist optics is planned for future development.

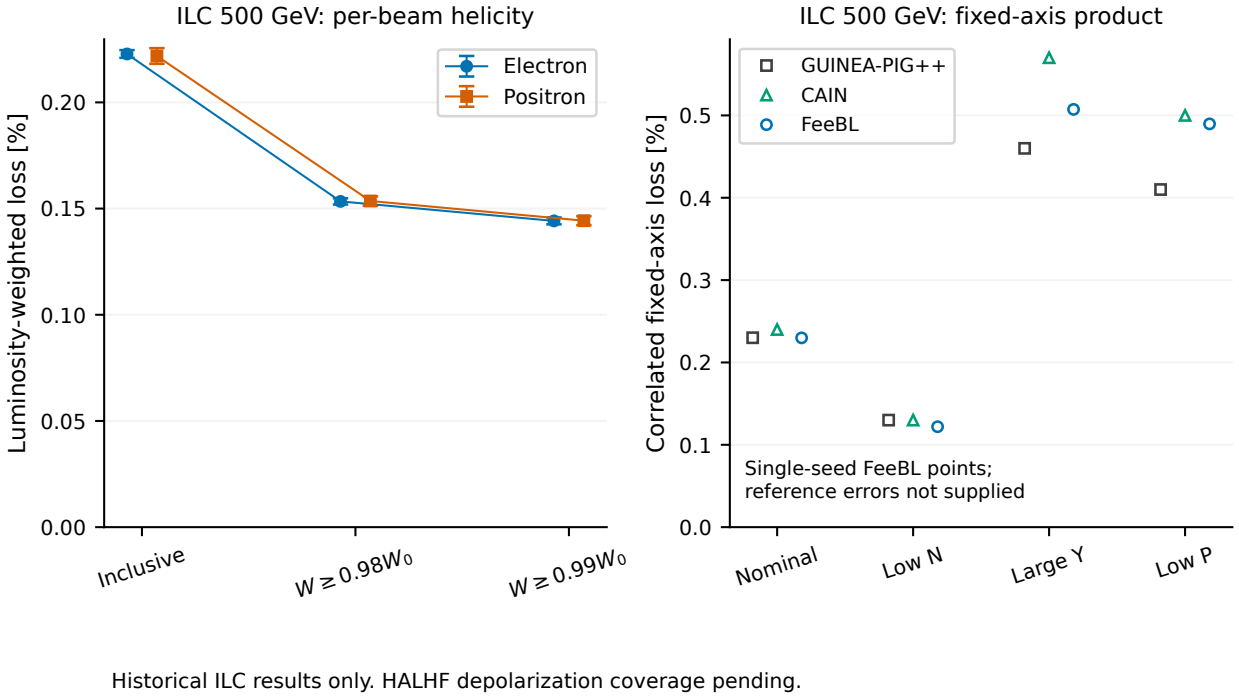


Figure 10: ILC 500 GeV depolarization. Left: luminosity-weighted local-helicity loss for each initially fully polarized beam, inclusively and with the 98% and 99% collision-energy selections. Capped errors are three-seed SEMs at fixed numerical settings; numerical shifts are separate. Right: inclusive correlated fixed-axis spin-product losses for the four 2009 optical configurations [5]. These retained FeeBL points have one seed each; their independent-seed errors have not been measured. The two panels use distinct observables and are not a measurement of conventional effective polarization.

The reference papers demonstrate physical and inter-code checks but do not document a complete mesh/slice/time/particle/seed convergence sequence for these rows. This limits a precision comparison; it does not imply their codes were never tested. The present optical variants likewise need independent seeds and domain studies. Observable matching and numerical refinement must precede a calibrated equivalence statement.

5 Horizontal-dispersion monochromatization

Direct resonant $e^+e^- \to H$ production requires collision energies near 125 GeV, rather than the 240–250 GeV Higgsstrahlung energy. A natural Higgs full width near 4.1 MeV motivates a machine response of comparable or smaller width [33, 34]. Narrowing alone does not establish a useful measurement: the luminosity, ISR, calibration, acceptance and backgrounds determine the available information. A single rate cannot separately determine an unknown signal normalization and intrinsic width.

We study horizontal dispersion at the interaction point. It correlates particle energy with transverse position so that spatial overlap preferentially selects compensating energy deviations [35]. It need not reduce either incoming beam's energy spread. The collision studies include beam–beam forces and beamstrahlung but disable ISR, spin transport and secondary backgrounds. Their results are pre-ISR machine spectra.

5.1 Gaussian mechanism and peak-response definition

Let independent fractional energy deviations have variances s_i^2 and $x_i = x_{\beta i} + D_i\delta_i$, with independent betatron sizes b_i. For head-on Gaussian beams, the overlap weights energy pairs by

$$\exp\left[-\frac{(D_1\delta_1 - D_2\delta_2)^2}{2B}\right], \qquad B = b_1^2 + b_2^2, \tag{17}$$

giving the collision-weighted variance of their fractional-energy sum,

$$V_{\rm coll} = \frac{(s_1^2+s_2^2)B + s_1^2 s_2^2 (D_1+D_2)^2}{B + D_1^2 s_1^2 + D_2^2 s_2^2}, \qquad \Lambda_{\rm ideal} = \sqrt{\frac{s_1^2+s_2^2}{V_{\rm coll}}}. \tag{18}$$

The first-order energy width is $W_0\sqrt{V_{\rm coll}}/2$. At fixed optics, the luminosity ratio is

$$\frac{\mathcal{L}(D_1,D_2)}{\mathcal{L}(0,0)} = \sqrt{\frac{B}{B + D_1^2 s_1^2 + D_2^2 s_2^2}}. \tag{19}$$

For equal beams and opposite dispersions this is $1/\Lambda_{\rm ideal}$. Equal same-sign dispersions leave the energy-sum width unchanged when the two input energy spreads are equal. These expressions omit radiation, disruption, crossing, hourglass and non-Gaussian correlations.

Force-free geometric controls at $W_0 = 250\,\text{GeV}$, $|D| = 0.35$ mm and $s_i = 0.0015$ give an opposite-sign width ratio 0.704534 versus the predicted 0.700608, a +0.5604% residual. The same-sign ratio is 1.003402 versus unity. Luminosity ratios are 0.702685 and 0.697154 versus 0.700608. These one-seed controls use artificially large beta functions to suppress divergence and hourglass effects. They establish geometric consistency at those settings, not an operating optics or a converged radiative result.

The peak response follows the beta-plus-delta model convolved with a Gaussian in Ref. [36]. With $u = W/W_\star$,

$$p(u) = fG(u-1;\sigma) + \frac{1-f}{\mathrm{B}(\alpha,\beta)}\int_0^1 x^{\alpha-1}(1-x)^{\beta-1}G(u-x;\sigma)\,dx, \tag{20}$$

$$G(v;\sigma) = \frac{e^{-v^2/(2\sigma^2)}}{\sqrt{2\pi}\sigma}, \qquad \sigma_G = W_\star\sigma. \tag{21}$$

The energy density is $p(W/W_\star)/W_\star$. The endpoint fraction f, beta-tail parameters and endpoint are fitted; σ_G is the common Gaussian response. It is neither the whole-spectrum RMS nor the RMS after a 98% energy cut. The latter selection applies independently to diagnostic energy and polarization means.

The model is integrated over the recorded bins. For normalized weights p_i, bin widths h_i and component probabilities g_i, t_i, the descriptive density objective is

$$J = \frac{\sum_i [p_i - A_g g_i - A_t t_i]^2 / h_i}{\sum_i p_i^2 / h_i}, \qquad A_g, A_t \geq 0. \tag{22}$$

Amplitudes are profiled by nonnegative least squares. This is not a Poisson likelihood or chi-square: repeated luminosity encounters are correlated. Numerical admission tests examine support, occupied bins, width relative to bin size, boundary solutions, quadrature, competing minima and nuisance profiles. Statistical errors on study results come from independent seeds, not the curvature of J.

The viewer's incoming-width factor $\sigma_{W,\mathrm{in}}/\widehat{\sigma}_W$ is distinguished from a measured control ratio $\widehat{\sigma}_W(D=0)/\widehat{\sigma}_W(D)$. The latter requires a matched zero-dispersion run. It cannot be inferred by replacing that run with the input Gaussian width. Frame estimates can fail resolution checks even when the integrated spectrum is adequate.

5.2 125 GeV incoming families and luminosity-retaining points

The ILC/LCF and HALHF families retain published 250 GeV normalized emittances, populations, relative energy spreads, lengths and rates while changing energies [37, 20]. Table 15 defines these exploratory low-energy assumptions. They are not a claim that the corresponding sources and final focus can deliver the stated 125 GeV collision. The LCF interpretation doubles only the crossing rate [38], so it has the same per-crossing spectrum as ILC.

Table 15: 125 GeV linear-collider input families. Ordered entries identify electron and positron; a single value applies to both. Luminosities are per IP.

Quantity	ILC / LCF	HALHF
Beam energy [GeV]	62.501531	187.5, 20.833333
Population [10^{10}]	2	1, 3
Normalized $\epsilon_{n,x}$ [μm]	5	90, 10
Normalized $\epsilon_{n,y}$ [nm]	35	320, 35
Reference β_x^*, β_y^* [mm]	13, 0.41	3.3, 0.10
Bunch length [μm]	300	150
Relative energy spread [%]	0.188, 0.150	0.150
Full crossing angle [mrad]	14	0 (assumed)
Ideal RF-like tilt	on	off
Vertical waist displacement [μm]	0	75
Crossings/s	6560 / 13120	16000

The ideal ILC crab map supplies perfectly prepared spatial tilt with its canonical energy companion. It does not model RF curvature, phase noise or cavity transport. Physical crossing still enters the invariant mass. The HALHF head-on geometry and 9:1 energy asymmetry are explicit assumptions; Gaussian moments cannot reproduce unprovided plasma-beam correlations. Initial longitudinal admission uses $-3\sigma_z \leq z < 3\sigma_z$ from fixed generation attempts. Surviving macroparticles retain their original weights; charge is not renormalized after truncation.

The initial scan contains sixteen linear-collider discovery points, varying horizontal focusing and dispersion with matched zero/same-sign controls. The selected points reduce β_x by four and apply $(D_1, D_2) = (+0.3712, -0.3712)$ mm for ILC/LCF or $(+0.5194, -0.5194)$ mm for HALHF. Vertical beta remains unchanged. The geometric expectations are a factor-two narrowing and unchanged luminosity; disruption, radiation and waist effects determine the actual result. A frozen selection rule is tested with three fresh paired seeds for each family; discovery samples do not enter the confirmation means. The rule retains at least 90% of matched-baseline luminosity and ranks admitted Gaussian-response widths. Both baseline and candidate must pass the response checks.

Table 16: Independent confirmation of the selected 125 GeV points. Errors are SEMs across three seed pairs. Ratio means retain the variation and covariance of both numerator and baseline.

Quantity	ILC/LCF	HALHF
Baseline σ_G [MeV]	150.1 ± 1.1	133.8 ± 1.1
Selected σ_G [MeV]	76.50 ± 0.35	67.21 ± 0.31
Paired width ratio	0.5096 ± 0.0015	0.5023 ± 0.0019
Paired luminosity ratio	1.023 ± 0.023	1.0010 ± 0.0031
Pre-ISR resonance-proxy gain	1.779 ± 0.023	1.8466 ± 0.0099

An interval-width ranking can suggest a further test when the fit fails, but is not substituted for an identified response width.

All twelve confirmation fits pass their stated response checks. Approximate 95% Student-t intervals for luminosity retention are [0.9241,1.122] and [0.9875,1.015], above the declared 0.90 screen. These intervals describe sampling at the tested resolution, not numerical or optics uncertainty. The selected responses remain much broader than the Higgs width.

Figures 11 and 12 show the redistribution into the peak under the selected settings and their matched baselines. The rate-relevant summary used here is the pre-ISR spectral overlap

$$K_H(W) = \frac{(\Gamma_H/2)^2}{(W - m_H)^2 + (\Gamma_H/2)^2}, \qquad R_H = \int K_H(W) f(W)\, dW, \qquad \mathcal{L}_{H,\mathrm{proxy}} = \mathcal{L} R_H, \quad (23)$$

where $f(W)$ is the normalized luminosity spectra and we use $m_H = 125\,\mathrm{GeV}$ and $\Gamma_H = 4.1\,\mathrm{MeV}$. R_H increases from 1.150% to 2.000% for ILC and 1.225% to 2.259% for HALHF. The proxy incorporates the luminosity cost, but is not a predicted signal yield or coupling sensitivity. ISR and process normalization are absent.

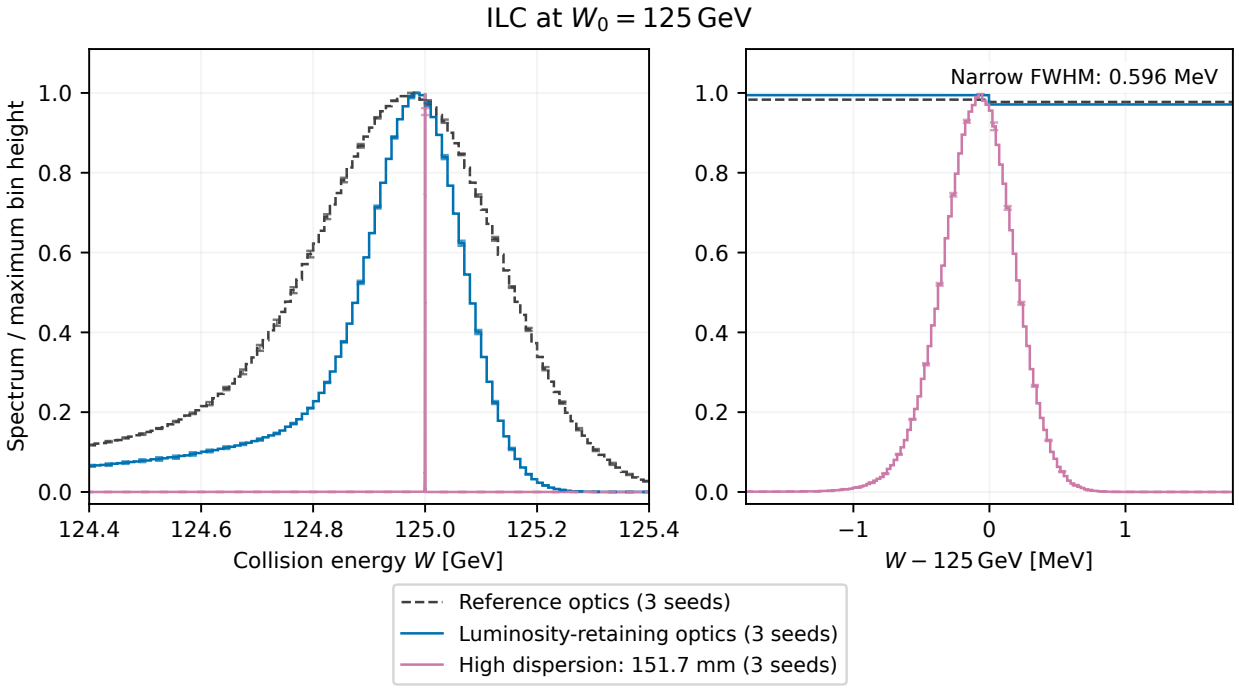


Figure 11: ILC confirmation spectra at 125 GeV for the reference and quarter-beta opposite-dispersion settings. Each realization is normalized over its complete spectrum before the peak is displayed. Capped bars are three-seed SEMs. The curves compare the two optical settings; they are not a reference-agreement test. LCF changes only the rate.

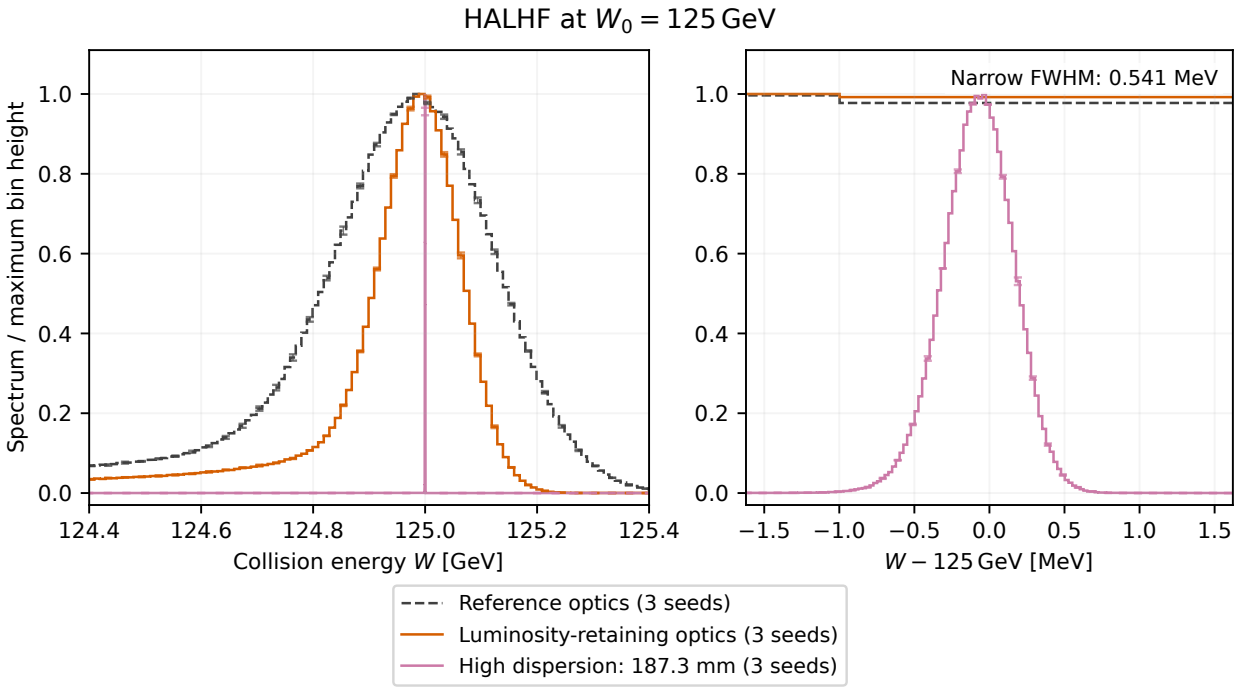


Figure 12: HALHF confirmation spectra with fixed 9:1 energy ratio and assumed head-on geometry. Normalization and error conventions follow Fig. 11. A narrower high-energy response coexists with the broad radiative tail outside the displayed interval.

5.3 High dispersion without a luminosity floor

The second scan asks how far dispersion must increase to approach a 1–4 MeV peak width. It holds the selected beta functions fixed at 3.25 mm for ILC/LCF and 0.825 mm for HALHF, while removing the luminosity retention requirement. The control for each point is now a zero-dispersion run at these same beta functions, not the original larger-beta reference. The head-on guide is

$$\sigma_{W,\mathrm{lin}}(D) = \frac{W_0\sqrt{s_1^2+s_2^2}}{2\sqrt{1+D^2(s_1^2+s_2^2)/B}}, \qquad \frac{\mathcal{L}_{\mathrm{geom}}(D)}{\mathcal{L}_{\mathrm{geom}}(0)} = \frac{1}{\sqrt{1+D^2(s_1^2+s_2^2)/B}}. \tag{24}$$

Canonical longitudinal reslicing is enabled throughout this scan. An incoming-map consistency correction is tested by 184 focused unit checks and 37 built-simulation checks; a large dispersion cannot be modeled by moving x while leaving slice scheduling attached to old ζ values.

The beta–Gaussian extraction at the MeV scale does not pass complete numerical qualification. Endpoint integration and differentiation were improved, and six noise-free fits recover their specified widths. However, expanded-family nuisance profiles in one Gaussian-limit test exhaust their evaluation bound, and some scalar optimizations fail an independent stationarity check. No fitted σ_G from this high-dispersion study is therefore admitted as a final measurement.

Instead, Table 17 reports the observed FWHM. Bin weights are divided by bin width, and the half-height crossings are interpolated between bin centers. The original 0.2 MeV histogram is too sparse for broad peaks: its largest noise fluctuation can split the half-height support and give an artificially narrow width. Conversely, grouping narrow peaks into 0.8 MeV bins produces appreciable broadening even for a smooth Gaussian. These effects require different analysis resolutions.

We reanalyse the saved spectra with one fixed resolution per optics class: 6.4 MeV for broad zero-dispersion and FCC-ee spectra, and 0.8, 0.4 and 0.2 MeV for the scan's respective 4, 2 and 1 MeV Gaussian-width targets. This policy is an analysis revision based on the diagnosed binning problem, not a new independent simulation. Every bin alignment at the chosen and twice-chosen bin width is checked. The observed peak must remain connected, and the resulting FWHM envelope

must lie within 5% of the primary value. Exact bin-integrated Gaussian controls quantify the finite-bin response; they do not subtract a fitted correction from the data. The original fine-bin results remain available. This resolves the histogram advisory at the tested resolution, while leaving particle, mesh and time convergence separate. An observed FWHM need not equal $2.355\sigma_G$.

Table 17: Fixed-optics horizontal-dispersion scan at $W_0 = 125$ GeV before ISR. Luminosity is in 10^{32} cm^{-2}s^{-1} per IP. Quoted uncertainties are standard errors across three independent collision seeds; rows without uncertainties use one seed. The final column, resolution, gives the largest absolute fractional FWHM change over individual seeds and bin alignments. The ILC/LCF luminosity uses the ILC 5 Hz rate not the 10 Hz LCF rate.

Family	$\lvert D\rvert$ [mm]	Seeds	$\mathcal{L}/10^{32}$	FWHM [MeV]	Resolution [%]
ILC/LCF	0.000	3	93.2 ± 5.9	424.4 ± 3.5	0.47
ILC/LCF	8.051	1	1.653	9.630	3.14
ILC/LCF	16.107	1	0.726	4.871	1.93
ILC/LCF	32.216	3	0.34130 ± 0.00043	2.5118 ± 0.0077	1.80
ILC/LCF	75.865	3	0.13850 ± 0.00035	1.0695 ± 0.0042	1.88
ILC/LCF	151.731	3	0.068441 ± 0.000052	0.5964 ± 0.0017	1.40
HALHF	0.000	3	144.72 ± 0.53	350.3 ± 1.4	0.77
HALHF	9.936	1	1.981	9.421	4.31
HALHF	19.878	1	0.951	4.671	3.10
HALHF	39.759	3	0.4694 ± 0.0020	2.433 ± 0.027	3.18
HALHF	93.629	3	0.19395 ± 0.00052	1.0276 ± 0.0058	1.57
HALHF	187.258	3	0.093141 ± 0.000070	0.5405 ± 0.0060	2.28

At the narrowest points, paired luminosity retentions are $0.369 \pm 0.023\%$ for ILC and $0.32436 \pm 0.00039\%$ for HALHF. The ILC control's 6.3% luminosity SEM dominates its retention uncertainty. Doubling only horizontal cells from 2048 to 4096 changes matched-seed FWHM values from 2.5007 to 2.4055 MeV and 2.4862 to 2.4119 MeV, shifts of -3.81% and -2.99%. Luminosity changes by $+0.223\%$ and $+0.772\%$. Thus MeV-scale observed narrowing is plausible, but representation and mesh effects exceed the sampling SEMs. The results do not establish a limiting physical peak width or a feasible final focus.

5.4 FCC-ee with its finite-angle crab-waist scheme

FCC-ee retains a 30 mrad full crossing angle and a 90% local crab-waist map. Crab-waist sextupoles orient the vertical beta-function waist along the opposing beam trajectory; this is distinct from RF cavities rotating the bunch envelope toward head-on overlap. RF bunch rotation is a separate proposal, not assumed in the FCC-ee baseline here [39].

The incoming moments are the published MonochroM ZH2IP values including equilibrium beamstrahlung [39]: $N = 5.54 \times 10^{10}$, geometric emittances $(3.56\,\text{nm}, 3.42\,\text{pm})$, $(\beta_x^*, \beta_y^*) = (90, 1)$ mm, $\sigma_\delta = 6.4 \times 10^{-4}$ and $\sigma_z = 4.74$ mm. The per-IP rate is $3.95330714 \times 10^7\,\text{s}^{-1}$ from 12 000 bunches and 91 km circumference. The beam energy is retuned to $62.5/\cos(15\,\text{mrad}) = 62.50703191$ GeV to center W_0 at 125 GeV. The source's 105 mm dispersion point is context; the executed scan is limited to 100 mm with the other incoming moments held fixed.

The local canonical waist map contains both the vertical displacement and its conjugate horizontal kick. It is not a transported model of the full sextupole system. Force-free comparisons with independent Gaussian overlap and vertical hourglass/waist weighting differ by -3.08% at zero dispersion and -2.54% at 100 mm. They satisfy the declared 15% coarse mechanism tolerance, not a percent-level accuracy requirement.

Figure 13 compares luminosity and observed widths. The luminosity decreases to 48.3% of the same-optics zero-dispersion realization at 100 mm. With the revised broad-peak resolution, the FWHM decreases from 130.93 to 81.14 MeV. All tested bin alignments retain one half-height

Table 18: FCC-ee radiation-on finite-angle crab-waist scan. Each row has 10 000 macroparticles per beam, one seed, 512×4096 transverse cells, 32 slices and two encounter subdivisions. Luminosity is per IP. Observed widths use 6.4 MeV bins. The range covers all alignments at 6.4 and 12.8 MeV; it is a numerical envelope, not a statistical error. No fitted Gaussian-response width is inferred.

$\lvert D\rvert$ [mm]	$\mathcal{L}$ [10^{34} cm^{-2} s^{-1}]	FWHM [MeV]	Resolution range [MeV]
0	20.099	130.93	[130.00, 132.93]
25	18.453	128.56	[127.22, 130.90]
50	15.052	109.84	[108.32, 111.93]
75	12.004	94.28	[93.41, 97.08]
100	9.714	81.14	[79.66, 83.99]

component, unlike the fragmented fine-bin estimates. The result is a stable description of these saved histograms, not a sampling or accelerator-convergence result: each FCC-ee point still has one seed. None demonstrates 1–4 MeV resolution. Larger particle, slice and time settings remain necessary for a quantitative collider prediction. Changing dispersion while keeping incoming equilibrium moments fixed also does not recompute storage-ring equilibrium, dynamic acceptance, lifetime or multi-turn stability.

5.5 Baseline-only analytical projections

The analytical projection in Fig. 14 uses the same rigid Gaussian-overlap approximation for three stated reference schemes: ILC at 14 mrad with ideal spatial tilt, the head-on HALHF candidate and FCC-ee at 30 mrad without RF tilt. "Rigid" means the bunch densities retain their assigned shapes while crossing; disruption and radiation do not evolve them. The physical crossing angle still contributes to the invariant energy even when the spatial overlap is ideally compensated. The curves retain source reference optics, without assigning one collider an additional focusing or bunch-population advantage.

For equal un-tilted beams at half-angle ϑ, a useful special case is

$$A = \epsilon_x \beta_x^* + \sigma_z^2 \tan^2\vartheta, \qquad \lambda^2 = 1 + D^2\sigma_\delta^2/A, \tag{25}$$

$$\mathcal{L}_{\text{rigid}} = \frac{fN^2}{4\pi\sqrt{A + D^2\sigma_\delta^2}\sqrt{\epsilon_y\beta_y^*}}, \qquad \sigma_{W,\text{lin}}^2 = 2E_0^2\left[\frac{\sigma_\delta^2\cos^2\vartheta}{\lambda^2} + \frac{\epsilon_x}{\beta_x^*}\sin^2\vartheta\right]. \tag{26}$$

SI lengths give luminosity in m^{-2}s^{-1}; divide by 10^4 for cm^{-2}s^{-1}. The first term narrows through energy selection while the uncorrelated angular term remains. The more general covariance calculation supplies the unequal-beam and ideally tilted curves. It omits hourglass/crab-waist weighting, canonical transport correlations and equilibrium changes. It is not a continuation of the FCC-ee simulated peak or a proof of realizable optics. At 100 mm the fixed-input model predicts a luminosity ratio of 0.753, whereas the native calculation gives 0.483. This discrepancy limits use of the projection as a continuation of the native scan; it does not disprove a model that deliberately omits the native dynamics. Matched force-free, hourglass, waist and radiation controls are required to isolate its origin.

For the fixed FCC-ee moments, the analytical Gaussian FWHM reaches 4, 2 and 1 MeV at approximately 3.86, 8.03 and 19.47 m dispersion. These metre-scale values are extrapolations, far beyond the 100 mm executed range. The corresponding model's angular floor is 0.621 MeV in Gaussian FWHM units, not a universal lower limit on a transported asymmetric peak. Exact-invariant quadrature provides a separate diagnostic of the linear approximation; faint curve segments mark an RMS difference above 5%. The exact RMS is not converted into a replacement peak FWHM.

Ideal spatial tilt is a limiting model, not a measured RF performance. ILC cavity amplitude and timing tolerances [40] cannot be converted into one fixed compensation fraction without their

distributions, correlations and transport. Likewise, a local FCC-ee waist map does not validate the full sextupole optics. These limitations set the scope of the horizontal-dispersion results. Vertical dispersion, longitudinal dispersion with energy–position correlations, chromatic waist matching and combined schemes [41, 42, 39, 43] remain future studies, to be identified separately in subsequent papers.

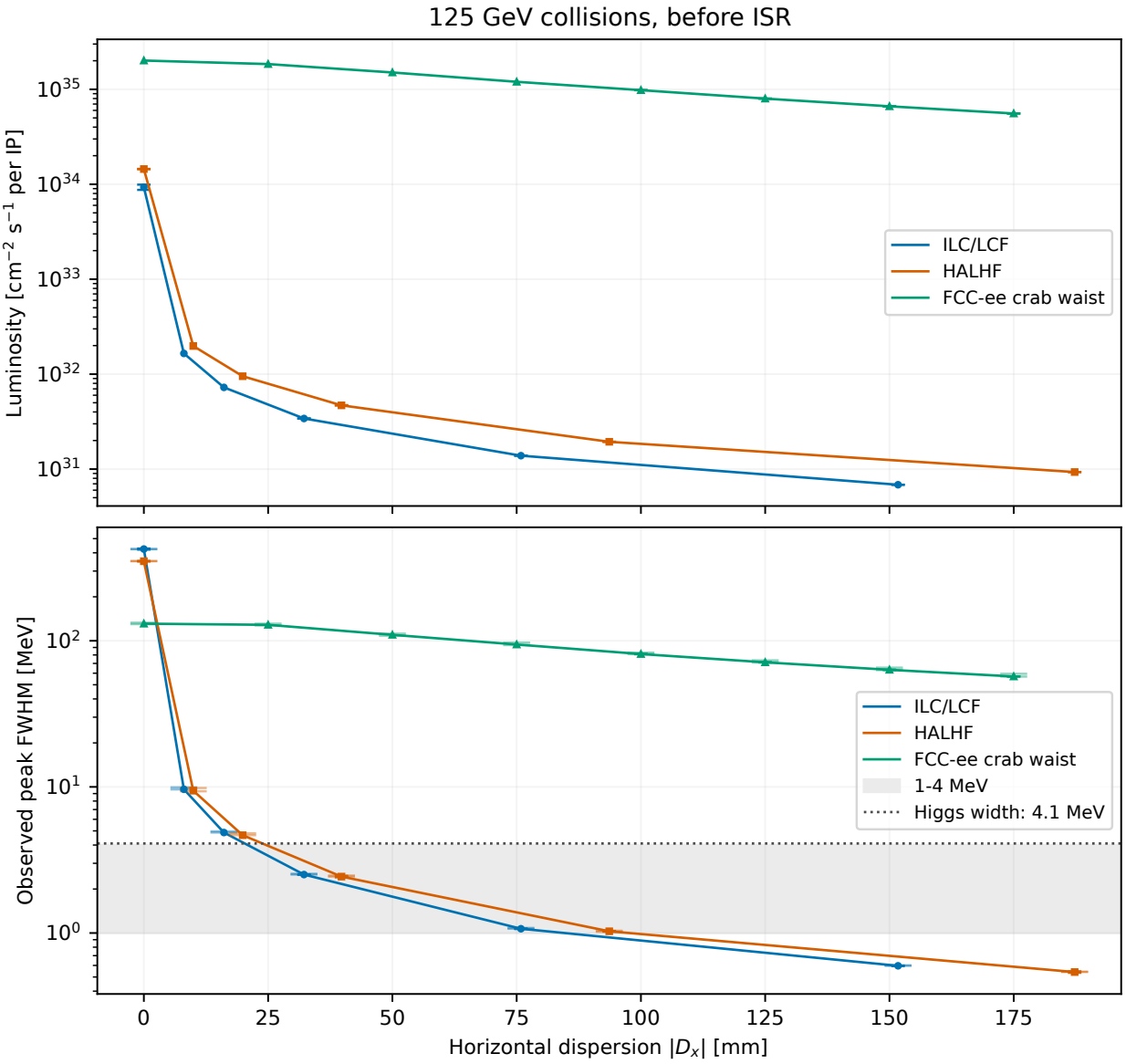
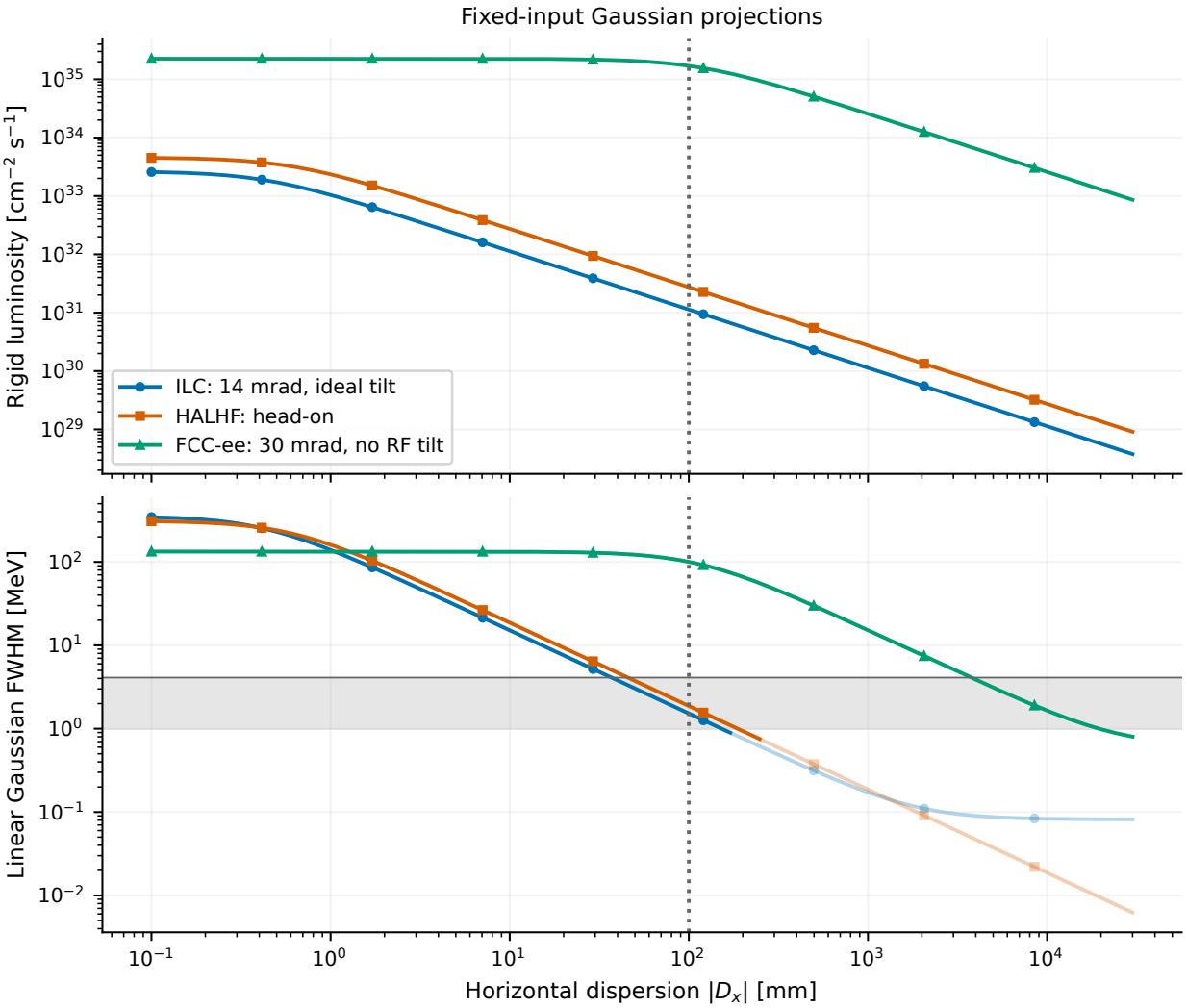


Figure 13: Collision luminosity and observed FWHM versus opposite horizontal dispersion at 125 GeV before ISR. Filled symbols have three seeds and capped SEM bars; hollow symbols have one seed. Separate pale width ranges show bin-alignment and analysis-resolution sensitivity, not statistical errors. Endpoint mesh controls are displayed separately. FCC-ee retains the source-based 30 mrad crab-waist geometry, with dispersion at most 100 mm. All three collider families are included in both panels. The band is 1–4 MeV and the line is $\Gamma_H = 4.1\,\text{MeV}$. These are different collider input optics; the points do not establish a feasible final focus or a limiting physical energy resolution.



Omitted: radiation, disruption, RF errors, transported optics and waist/hourglass evolution.
FCC-ee L(100 mm)/L(0): rigid model 0.753; native 0.483. Matched controls are required.
Faint width segments: nonlinear RMS check differs by >5%; beyond 100 mm is projection only.

Figure 14: Baseline-only analytical projections on a logarithmic dispersion axis. ILC uses 14 mrad with ideal spatial tilt, HALHF the head-on candidate, and FCC-ee 30 mrad without RF tilt. All curves use the same rigid Gaussian model and their reference optics. The width is the first-order Gaussian FWHM, not a fitted observed peak. Finite RF errors, transported optics, hourglass/crab-waist weighting, radiation, disruption and equilibrium adjustment are omitted. Faint segments flag a greater than 5% nonlinear kinematic diagnostic. The vertical line is the FCC-ee simulation limit; larger dispersions are analytical projections only.

6 Program and data release

The GitHub release accompanying this contains the source code, runnable cards, example inputs and outputs, digitized reference tables, and analysis scripts [44].

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